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Real-time policy for yard allocation of transshipment containers in a terminal

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ABSTRACT

In this article, we investigate the problem of allocating storage space in a container terminal's yard to transshipment containers. The main decision here concerns the block to which a container is assigned for storage until it is loaded later by another vessel. We propose a setting where some target performance measures are imposed on the discharge operations. In turn, the allocation decisions are made so as to reduce driving time from the storage blocks to the berth locations of the vessels that will pick up the containers. The trick here is to find an appropriate trade-off between the times spent on discharge and loading so that neither are delayed significantly. Using results from renewal theory, queuing theory and machine learning, we are able to quantify the effect of our allocation decisions on quay crane productivity. Thereafter, we formulate a mathematical optimization problem for the yard-allocation of containers and apply a meta-heuristic to solve it. Our method was developed for deployment by a software consultancy company for container terminals. We test our method using a real-time simulation and compare it with a benchmark from the literature. We show that our method generates reductions in vessel berth times and provide an overview on its economic impact.

1. Introduction

With trade volumes demonstrating significant growth in recent years due to factors such as globalization and the set up of new economic centers that have shifted the gravity of international economics throughout the years (Surugiu and Surugiu, 2015), maritime traffic has witnessed comparable growth (Ducruet and Notteboom, 2012). This growth in maritime traffic calls for sophisticated measures that strive to maximize the efficiency of the logistics involved in a wide range of operations, from vessel berth scheduling to the routing of terminal container carriers.

Our research is inspired by global demand of container terminals to increase their service efficiency of carriers, especially vessels. Increasing the service efficiency is characterized by minimizing the berth times of carriers for the same call. This is becoming more important with increasing vessel and call sizes (Notteboom and Vernimmen, 2009). We pay particular attention to container handling operations where we focus on allocating the containers to storage blocks in the yard. This invokes our main research question, "Can better optimization of yard-space allocation of containers lead to notable reductions in carrier berth times?".

The problem discussed here focuses on vessels only, although it could be trivially extended to other modes of transportation. To that end, our research concerns transshipment operations, common to ports like Singapore (Wee Kwan Tan and Hilmola, 2012). For vessels, the berth time is largely determined by the work queues of the quay cranes (QC)s. The processing of the work queues, in turn, depends on the availability of the transportation serving the quay cranes. This availability is directly determined by three

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main factors. The number of available transport units, the units' driving times and waiting times at the yard. Zhen (2016) states that yard traffic congestion has become the bottleneck that limits a port's efficiency.

Transportation in a terminal can be done with several types of equipment, like straddle carriers, automatic guided vehicles (AGVs) and terminal trucks which are often in a limited availability per crane due to associated capital and operational expenditure. The second factor refers to the time a truck spends driving between the quay crane and the yard. The containers are transported from and to a storage block in the yard. If the storage block is distant from the berth location of the vessel, the trucks will take more time to deliver the container and drive back again to the quay crane to transport the next container. This may cause a quay crane to wait until a truck arrives, thereby delaying it and the associated vessel. A similar logic holds for loading. The last factor represents the time that a truck has to spend in the yard awaiting service by the yard crane. In principle, the more containers the yard crane has to handle in a given period of time, the busier it will be, possibly delaying trucks that arrive later at its service queue, and thus, the quay cranes they serve.

Our objective is to devise a real-time block-allocation policy by which the berth times of vessels are minimized. The real-time component refers to the fact that the block allocation can change during the unloading of a ship due to changing workload at the yard. Dekker et al. (2007) performed a container stacking study and found out that fixed block allocations may lead to many short-lasting yard crane work overloads, hence the need for real-time policies. This is in contrast to existing static approaches that consider fixed allocations while disregarding the dynamics of QC productivity.

As for berth-time minimization, we strive to accommodate the trade-off between discharge and loading operations by placing the containers as close as possible to their future loading locations – assumed to be known – without significantly delaying discharge. This is done with the help of a mathematical optimization model that incorporates the effect of an allocation decision on the QC productivity in real-time – through the aforementioned factors – before deciding on the final allocation. To that end, we propose a mechanism to estimate the QC productivity in real-time, using results from queuing theory, renewal theory and machine learning. We integrate the prediction mechanism in our optimization model.

Our problem was firstly discussed in Zhang et al. (2003) which serves as the benchmark of our research. The authors devise a method by which they minimize the time spent on discharge by a vessel with no regards to the real-time impact of their decision nor to loading. Through this comparison, we highlight the virtues of balancing loading and discharge alongside accurate real-time calculation of performance measures resultant to allocation decisions. Many papers have appeared since on this problem, but only few considered congestion like Zhen (2016) and Feng et al. (2022) while our real-time approach is new.

Our study is carried out in consultation with a renowned consultancy bureau for container terminals and ports with the intention of deploying it in future projects.

The rest of this paper is organized as follows. Section 2 discusses the literature review and our contributions. Section 3 describes the problem in greater detail. Section 4 presents the relevant mathematical optimization model assuming given QC productivity estimates. Section 5 discusses the derivation of QC productivity estimates. Section 6 analyzes the numerical experiments through which we assess and compare our policy to that of Zhang et al. (2003). Lastly, Section 7 asserts the conclusions.

2. Literature review and contributions

The optimization of terminal operations has been a major research topic in the last few decades. There are many operations that take place, with different operations invoking different optimization problems. Several reviews have appeared on the topic viz. Vis and De Koster (2003), Carlo et al. (2014) and Weerasinghe et al. (2023).

Container terminal operations can be split up into berth operations (berth allocation and quay crane operations), storage yard operations and gate operations with land transportation. These operations can be considered at a design level, a tactical planning level and an operational scheduling level. For example, crane scheduling where work queues are planned for cranes in order to enhance their productivity has been extensively discussed in academic literature (Bierwirth and Meisel, 2010).

In this paper we focus on the allocation of transshipment containers to storage in the yard. The yard is the main decoupling point between landside and seaside operations where containers are stored until they leave the terminal. The yard consists of several blocks with dedicated yard cranes, sometimes parallel or perpendicular to the berth. The main challenge concerns where to store a container, such that associated equipment (like trucks and yard cranes) could be scheduled without causing significant delays to the QCs.

With regards to yard space allocation for containers, a lot of literature focuses on the grounding decision where a particular location within a block – defined by a bay, row and tier – has to be assigned given a certain block. This is a more detailed problem, whose input is determined by the output of our problem, and where the objective is often the minimization of handling moves during container retrieval. Kim et al. (2000) discusses this problem, taking into account the weight distribution of containers among the blocks, while Kim and Kim (1999) deals with various specifications of the vessel arrival rate. Other literature such as de Castillo and Daganzo (1993) and Kim (1997) focuses on simply evaluating the expected number of rehandles associated with a certain grounding policy. Dekker et al. (2007) and Borgman et al. (2010) provide comprehensive simulation studies of location assignment within a block. They also find that fixed block allocation policies may lead to many short-lasting (15 min) overloads of stacking cranes. These overloads will cause delays in handling trucks, with the consequence that quay cranes do not receive trucks in time. Hence real-time block allocations should be considered. Zhen (2016) derives a formula for the expected travel time of vehicles to the yard, taking congestion into account. He later uses these as input for the yard planning on a tactical level. To solve this problem he uses a Squeaky-Wheel heuristic. The same problem was tackled by Feng et al. (2022) who propose a nonlinear model which is solved by heuristics based on the Frank–Wolfe algorithm and the subgradient optimization method. Both papers do not consider real-time

planning.

On a broader level, there are several papers (Taleb-Ibrahimi et al., 1993) integrates both resource planning and container handling in blocks. Lee et al. (2010) investigate different vehicle-dispatching algorithms with the objective of minimizing the quay crane's makespan. They determine the minimum storage space requirements and predict the number of rehandles associated with a given container, which they eventually strive to minimize. Other papers such as Holguin-Veras and Jara-Diaz (1999) deal with the pricing aspect of space allocation, where a price is charged for the utilization of a particular space depending on a set of predefined criteria.

Nonetheless, some research has been dedicated to an optimization framework such as Kim and Park (2003) who introduce a simple optimization model where the travel times are minimized subject to capacity constraints. More comprehensively, Han et al. (2008) study yard strategy optimization exclusively through yard template design, yard crane deployment and block assignment without focusing on QC productivity and travel times from the berths to the blocks. Jiang et al. (2012) extend on this by proposing dynamic yard templates through space-sharing strategies.

Recent research was done on the integration of quay crane scheduling and block allocation (Hu et al., 2019; Yue et al., 2023). They integrate the scheduling of the quay cranes with the block allocation, by defining a comprehensive MIP problem and solving it with heuristics. Such a deterministic approach ignores queuing effects and resulting short term work overloads, which are the focus of real-time scheduling.

Most notably, Zhang et al. (2003) consider other factors alongside the joint workload of both loading and unloading containers on yard cranes such as the travel distances of horizontal transportation between the quay and the yard. The authors consider a static setting with a relatively long planning horizon whereby the workloads of the yard cranes are firstly balanced followed by the minimization of driving distances.

To the best of our knowledge, little research has been dedicated to addressing the problem of block assignment in real-time in order to minimize the vessel berth times. Most proposed approaches make use of simulation studies as in Bruzzone and Signorile (1998) and Ding et al. (2023) rather than well-formulated optimization problems. Although these studies are useful in modeling the dynamics of container handling and approximating the real-life performance of a policy, the decisions involved are often made in an obscure 'black-box' fashion, which is not very informative from an analytical perspective. Not to mention, simulation results often lack robustness, disabling them from serving a general reliable framework.

Our proposed research aims to establish a methodology by which transshipment operations could be optimized in real-time, in contrast to long planning horizons such as in Zhang et al. (2003). Firstly, we aim to set up a model that explicitly incorporates real-time performance measures of the discharge process. This allows us to momentarily measure the impact of the decisions we make. We also take into account the resources advantages or constraints such as number of available transportation units. Disregarding the resources as in Zhang et al. (2003) may prevent the policy from exploiting the associated advantages. For example, the existence of a sufficient number of trucks could allow longer distances without delaying the vessel departure.

Additionally, Zhang et al. (2003) only optimize driving times during discharge while completely neglecting loading. We show that reductions in the berth times are attainable when the future loading locations of the containers are taken into account. The approach in Zhang et al. (2003) also relies heavily on some impractical assumptions such as knowledge of future vessel berth times, while our method neglects this assumption.

3. Problem description

Here, we provide a detailed description of our decision problem. Our problem is characterized by extensive mathematical notation which we summarize in the table in Appendix A. Consider a container terminal where vessels arrive to discharge and load containers. We assume that containers are transported by terminal trucks or Automatic Guided Vehicles (AGVs) to the yard and put by gantry cranes, either manual or automatic, into a stack. The terminal layout, on the other hand, can be horizontal or vertical. Ideally, the container terminal would like to minimize the vessel berth times (service times) as much as possible for the sake of efficiency. The idea is to be able to meet a target berth time with some probability α, or quantitatively expressed:

$$P_b \geq \alpha \tag{1}$$

with P_b being the probability that vessel berth b can be processed in the target time. The target time is determined by a multitude of factors related to the number of containers to be processed and contractual agreements with the liner shipper. The berth time of the vessels is largely determined by the time during which the quay cranes assigned to the vessel process their work queues. Therefore, maintaining a high crane productivity λ_q is of importance. Crane productivity is expressed in several ways, which we restrict to the rate by which container orders are processed in minutes. Taking crane productivity into account, (1) can be rewritten as follows:

$$\Pi_{q \in Q_b} P(\lambda_q) \geq \alpha \tag{2}$$

with Q_b being the set of quay cranes involved in berth b and $P(\lambda_q)$ is the probability that crane q finishes its work queue in the target time while operating at an average productivity λ_q. $P(\lambda_q)$ is determined through the distribution of the truck inter-arrival times at crane q. We will return to this issue in Section 4.1. The product form in (2) assumes independence among quay crane operations. In reality, this may not be the case as the decision to send containers to a certain block makes the block's yard crane less available for trucks from other quay cranes. Below, we demonstrate how we account for this dependency by estimating the waiting time at each block.

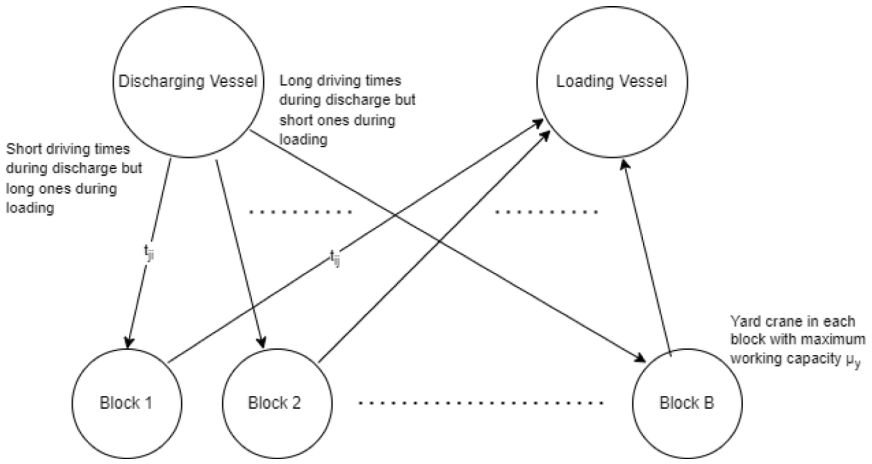


Fig. 1. Example of our problem illustration.

Our decision problem concerns the allocation of discharge containers to blocks in the yard such that quay crane productivity is maintained. We do not deal with the grounding decision or house-keeping moves.[1] The allocation decisions can affect the quay crane productivity as the trucks being dispatched to the allocated block may travel long distances — if the block is far. Furthermore, yard crane occupancy with other discharge and loading requests from other quay cranes — or even the same one, can elevate waiting times at the yard. When the trucks take too long before they return to their assigned quay crane, the crane might have to wait for them before discharging any containers off the vessel which reduces its productivity. On the other hand, if more trucks are available to dispatch containers, this can offset the delays resulting from long cycle times to and from the yard.

For loading containers, once the loading plan has been settled, there is little scope for optimization, as their positions in the yard were already determined during their unloading and trucks have to drive to their corresponding blocks to retrieve them. However, if we knew the berth location of the vessel that is going to pick up a container in the future, we might allocate it to a nearby block. On the other hand, this can be at the expense of the vessel discharging the container for the reasons described above. Therefore, one has to accommodate the trade-off between discharge operations and loading ones. Fig. 1 provides a graphical illustration of our problem.

To proceed further, we lay down a set of assumptions which we are going to need to develop a mathematical model that addresses our problem. We assume we have a discrete set of berth locations L along the quay. For each vessel berth, we know where the vessel will berth but not when. Additionally, we have a set of B storage blocks. For each block $i \in B$, we have one seaside yard crane while we disregard its storage capacities. According to Gharehgozli et al. (2015) the interaction between the seaside and landside crane in a block is only 1.88% of the time, hence we assume that the capacity of the seaside crane is not influenced by the landside crane. For each berth location $j \in L$ and block $i \in B$, we have fixed travel times t_{ji} and t_{ij}. We also associate each quay crane with a given berth location. For each quay crane, we have predetermined discharge work queues which have to be executed in sequence, and loading work queues with no particular ordering.[2]

For each quay crane q, we have a fixed number of trucks n_q that serve this crane only. Note that while many terminals work with pooling policies where trucks are shared among QCs, the analysis becomes increasingly complex and a closed form for the associated performance measures is unlikely. Each truck can carry at most one container at a time, and the containers are similar in a transportation context. For discharge containers, we only know the berth locations of the vessels that will pick them up in the future, but not the pick-up times. The discharge work queues also have to be finished completely before processing the loading work queues.

For both the quay and yard cranes, we assume stochastic service times. This is due to the different movements that they have to execute as they visit different sections in vessel bays and yard blocks. The service times follow some distribution with rates μ_q for quay cranes and μ_y for yard cranes, with $\mu_q > \mu_y$ since the quay crane is generally able to produce more moves per hour.

We also assume that the service discipline for trucks arriving at either crane types is First Come First Serve (FCFS) and that trucks cannot depart from a crane before their service is complete. The service times are independent. This also applies to truck movements. With these assumptions at hand, we present our mathematical model in the following section.

[1] Movement of a container from one block to another after it has been grounded earlier.

[2] In practice, the ordering of loading work queues is subject to some strict conditions, which we relax slightly for ease of analysis by assuming containers loaded in the same planning epoch are interchangeable in ordering.

4. Mathematical model

In this section, we present our mathematical model. Our approach consists of the following four steps in logical ordering: (1) Quantify the relationship between QC productivity and truck cycle times to and from a crane. (2) Use this quantification to estimate the QC productivity in real-time. (3) Define a mathematical optimization model for block allocation of containers. (4) Solve this model. The sequence of these four steps is illustrated in Fig. 2.

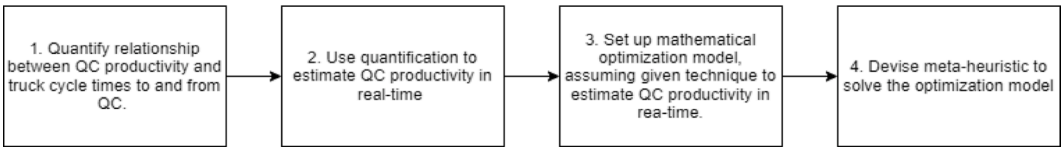


Fig. 2. Flowchart depicting the sequence of steps that compose our approach.

Steps 3 and 4 are more direct and treated in this section. Steps 1 and 2 are more complicated and treated in Section 5. It is important to be able to estimate the QC productivity at different intervals to be able to assess the impact of our block-allocation decisions and avoid serious delays. For now, we assume that we have a ready-made technique to estimate the QC productivity in real-time without loss of generality. We can then proceed with setting up our optimization model (Step 3) in Section 4.1 and propose a meta-heuristic to solve it (Step 4) in Section 4.2. In the sequel, we use the following notation next to the list in Table 1. With the $\hat{x}$ symbol we indicate an estimate, with a bar $\bar{x}$ an average and with x^* a target of a value x.

4.1. Optimization model

We consider a discrete-time setting where for each berth b and crane $q \in Q_b$ we have the decision epochs of constant duration $\{t_1^q, t_2^q, \dots, t_T^q\}$. Here, t_k^q represents the kth decision epoch during which crane q is discharging with the number of epochs being in accordance with the predefined discharge target time. Decision epoch t_k^q starts at real time $t^q(k)$, which satisfies:

$$t^q(k) = \mathbf{t}_a^q + \Delta(k-1) \tag{3}$$

for $k \in \{1, 2, \dots, \frac{\mathbf{t}_d^q - \mathbf{t}_a^q}{\Delta}\}$. Here $\mathbf{t}_a^q$ and $\mathbf{t}_d^q$ are the times during which discharge starts and ends for crane q - we assume that either times are multiples of Δ – with Δ being the length of a decision epoch – without loss of generality. Naturally, we have that $\mathbf{t}_a^q$ is assumed the same for $\forall q \in Q_b$, while $\mathbf{t}_d^q$ may vary depending on the workload. The utilization of discrete-time decision epochs serves as an approximation to the continuous time process. The quality of the approximation improves as Δ decreases. However, this comes at the expense of computational tractability.

In each decision epoch t_k^q, we assume that a certain number of containers from the discharge work queue of q has to be processed. The containers are distinguished by the berth locations where they will be picked up in the future, although this information may not always be known. However, common contractual agreements between shipping lines and terminal operators concerning dedicated use of quay space can often be determinant of ships' berthing locations as explained in Theofanis et al. (2009). That said, we have the parameter N_{qj}^t representing the number of containers that are discharged by crane q in epoch t and expected to be loaded at location j later in the future. The number of containers in period t discharged by crane q for which the future berth is unknown or have to leave through the landside are denoted by N_q^t.

For each of the $\sum_{j \in L} N_{qj}^t + N_q^t$ containers, a block-allocation decision is made. Our allocation decisions in epoch t result in a service rate λ_q^t at quay crane q as it has a limited set of trucks or AGVs only and the longer the trucks travel, the less containers crane q can handle. In particular, λ_q^t is difficult to determine precisely. In Section 5, we explain how an estimate $\hat{\lambda}_q^t$ of this quantity can be made which we use as a substitute in our optimization model instead.

A small remark on the fixed number of vehicles for horizontal transportation. We assume that this is fixed due to the associated operational expenditure of deploying more vehicles, which we assume the terminal is unwilling to put up with, even with unmanned AGVs that may require charging or deterioration costs. Our method, however, naturally extends to the case with varying vehicle numbers as the forecast $\hat{\lambda}_q^t$ is made per epoch, where different vehicle numbers can be used as input in different epochs.

Given the estimates $\hat{\lambda}_q^t$, we want to calculate the probability P_b that the process counting the number of containers discharged by each crane $q \in Q_b$ during its target discharge time makes at least C_q counts — where C_q is the total number of containers ought to be discharged by quay crane q. This process is identical to the process counting the number of truck arrivals at q.

Let $\lambda_q(s)$ be a function defining the truck arrival rate at q in time s, where s is continuous. $\lambda_q(s)$ is be defined as follows:

$$\lambda_q(s) = \begin{cases} \vdots \\ \lambda_q^t \\ \vdots \end{cases} \quad \Delta(t-1) \le s < \Delta t \tag{4}$$

with $t \in \{1, 2, \dots\}$ being an epoch number. Let $P(\lambda_q^{[1,\dots,T]})$ be the probability that the target berth time is met given arrival rates $\lambda_q^{[1,\dots,T]}$ representing the λ_q^t values in decision epochs $\{t_1^q, \dots, t_T^q\}$. We will elaborate on how this probability may be approximated in Section 5.2.2.

Let R_q^{ti} and R_{qj}^{ti} be decision variables describing the numbers of container discharged by q in period t and stored in block i with a known future loading location j for the latter, and $\hat{j}$ is the present berth location of the vessel upon which q operates. Additionally, let L_q^{ti} be a parameter input representing the loading orders from crane q at block i in period t. L_q^{ti} is input as we have no control over where the containers are loaded from once they are allocated.

Furthermore, let $\overline{t_q^t}$ and $\overline{w_q^t}$ represent estimates of the average driving time and waiting time of the trucks serving quay crane q resultant to the allocation decision in epoch t. Note that $\overline{w_q^t}$ requires an estimate of the average waiting time $\hat{w}_i^t$ at block i in decision epoch t. The average waiting time $\hat{w}_i^t$ is estimated as it is difficult to know the exact waiting time without knowing the order of the arrival of the trucks at the block. This is complex to keep track of, so we use an average as an approximation.

In Section 5.2.2, we explain how the estimate $\hat{w}_i^t$ can be obtained. For now, we simply assume it is a function $g(.)$ of the estimated arrival rate $\hat{\lambda}_i^t$ at block i in epoch t. We require that $\hat{\lambda}_i^t < \mu_y$ so that the service capacity of the yard crane is respected.

Given estimates $\overline{t_q^t}$ and $\overline{w_q^t}$, we can use them alongside n_q to estimate λ_q^t by means of a function $f(.)$:

$$\hat{\lambda}_q^t = f(n_q, \overline{t_q^t}, \overline{w_q^t}) \tag{5}$$

and is assumed to be constant throughout epoch t. We will describe a regression model for $f(.)$ in Section 5.2.3. With the above definitions, we can now define our optimization model for decision variables R_{qj}^{ti}, R_q^{ti}, $\hat{\lambda}_i^t$, $\hat{w}_i^t$, $\overline{t_q^t}$, $\overline{w_q^t}$ and $\hat{\lambda}_q^t$ as follows:

$$\min \sum_{q\in Q}\sum_{t=1}^{T}\sum_{i\in B}\sum_{j\in L} t_{ij} R_{qj}^{ti} \tag{6}$$

$$s.t. \quad \sum_{i\in B} R_{qj}^{ti} = N_{qj}^t \qquad 1 \le t \le T, \qquad \forall q \in Q, \forall j \in L \tag{7}$$

$$\sum_{i\in B} R_q^{ti} = N_q^t \qquad 1 \le t \le T, \forall q \in Q \tag{8}$$

$$\hat{\lambda}_i^t = \frac{\sum_{q\in Q}(R_q^{ti} + \sum_{j\in L} R_{qj}^{ti} + L_q^{ti})}{\Delta} \qquad 1 \le t \le T, \forall i \in B \tag{9}$$

$$\hat{\lambda}_i^t \le \mu_y - \epsilon \qquad 1 \le t \le T, \forall i \in B \tag{10}$$

$$\hat{w}_i^t = g(\hat{\lambda}_i^t) \qquad 1 \le t \le T, \forall i \in B \tag{11}$$

$$\overline{t_q^t} = \frac{\sum_{i\in B}(t_{\hat{j}i} + t_{i\hat{j}})(R_q^{ti} + \sum_{j\in L} R_{qj}^{ti})}{\sum_{j\in L} N_{qj}^t + N_q^t} \qquad 1 \le t \le T, \forall q \in Q \tag{12}$$

$$\overline{w_q^t} = \frac{\sum_{i\in B} \hat{w}_i^t(R_q^{ti} + \sum_{j\in L} R_{qj}^{ti})}{\sum_{j\in L} N_{qj}^t + N_q^t} \qquad 1 \le t \le T, \forall q \in Q \tag{13}$$

$$\hat{\lambda}_q^t = f(n_q, \overline{t_q^t}, \overline{w_q^t}) \qquad 1 \le t \le T, \forall q \in Q \tag{14}$$

$$\Pi_{q\in Q^b} P(\hat{\lambda}_q^{[1,\dots,T]}) \ge \alpha \qquad \forall b \in \mathcal{B} \tag{15}$$

$$R_{qj}^{ti}, R_q^{ti} \in \mathbb{Z}^+ \qquad \forall i \in B, 1 \le t \le T, \forall q \in Q, \forall j \in L \tag{16}$$

$$\hat{\lambda}_i^t, \hat{w}_i^t, \overline{t_q^t}, \overline{w_q^t}, \lambda_q^t \ge 0 \qquad \forall q \in Q, 1 \le t \le T, \forall i \in B \tag{17}$$

The objective function (6) minimizes the driving times during loading. Constraints (7) and (8) ensure that every discharged container gets assigned to a storage block. Constraint (9) implies that the arrival rate $\hat{\lambda}_i^t$ at block i in epoch t is estimated by the ratio between the total number of orders that the yard crane ought to handle within the epoch and the length of the epoch. This constraint reflects an important aspect of the problem; that loading orders may increase waiting times in the yard. Furthermore, $\hat{\lambda}_i^t$ must respect the block service capacity as enforced by constraint (10) for some small $\epsilon > 0$. Constraint (11) estimates the expected waiting time at a block. Constraints (12) and (13) estimate the average driving and waiting times of the trucks using weighted averages of the allocated containers from QC q in epoch t. These estimates are then used in Constraint (14) to estimate the QC productivity using $f(.)$. Constraint (15) ensures that the target time is met with a certain probability. This constraint, along with the objective function, jointly accommodate the trade-off between discharge and loading. Lastly, (16) and (17) specify the domains of the decision variables.[3]

Problem (6)–(17) is highly non-linear and, thus, difficult to solve using exact methods. In the following section, we propose an adjustment to the formulation and a fast meta-heuristic to tackle the associated computational challenges.

4.2. Solution approach

Our methodology is specifically driven by the concept of Receding Horizon Control (RHC), whose application in container terminal problems has shown to be very effective (Alessandri et al., 2007). RHC involves planning ahead for a limited time horizon as information is gradually revealed.

[3] $\mathbb{Z}^+$ is the set of non-negative integers.

Consequently, an adjustment has to be made to constraint set (15). Specifically, if there are $|Q_b|$ cranes involved in berth b, then a sufficient condition that the target discharge time is met with probability α is that each quay crane finishes its work queue in the target discharge time with probability $\sqrt[|Q_b|]{\alpha}$. This translates to finding a mean arrival rate λ_q throughout the discharge period that guarantees this.

Such a value of λ_q can be found using the bisection method described in Algorithm 1 in Appendix B and we refer to it as λ_q^*. Observe that the calculation of such an arrival rate is dependent on our knowledge of the length of the discharge period for crane q and the number of containers C_q in the work queue of q. We assume we know either quantities as these can be easily provided by the vessel operators beforehand.

The implication to determining a target λ_q is that constraint set (15) is replaced by:

$$\frac{\sum_{t=\max\{t_1^q,t_k^1\}}^{\min\{t_T^q,t_k^T\}} \hat{\lambda}_q^t}{\min\{t_T^q,t_k^T\}-\max\{t_1^q,t_k^1\}+1} \geq \lambda_q^* \qquad \forall q \in Q \tag{18}$$

where t_k^1 and t_k^T are the start and end epoch of the receding horizon. The advantage of this constraint formulation is that it allows us to partition the original problem into several smaller subproblems – concerning each planning horizon – that are much more computationally tractable, although it imposes unnecessarily high efficiency standards since we may be able to afford lower quay crane productivity levels at later time horizons.

As will be explained in Section 5, the λ_q^t values determined by $f(.)$ are underestimates of the actual QC productivity. Consequently, we replace (18) with:

$$\frac{\sum_{t=\max\{t_1^q,t_k^1\}}^{\min\{t_T^q,t_k^T\}} \hat{\lambda}_q^t}{\min\{t_T^q,t_k^T\}-\max\{t_1^q,t_k^1\}+1} \geq \lambda_q^* - d \qquad \forall q \in Q \tag{19}$$

where $d > 0$ is some small decrement that can be seen as a tolerance for inefficiency which encourages exploration of slightly less efficient allocations. We noticed that such a decrement leads to improved results. It is important that d is not too large, otherwise the target time may be severely violated as the heuristic accepts inefficient allocations. The choice of d can be determined through experimentation.

The meta-heuristic we propose to solve (6)–(17) is based on a neighborhood search. It strives to greedily improve the objective value at every iteration by picking a small population of the best found solutions and applying some perturbation to each of those solutions that aims to generate new feasible solutions with lower objective values. Thereafter, it selects the best found solutions again and repeats the procedure for a fixed number of iterations.

The heuristic starts with generating an initial solution $\mathcal{R}^1$. This is normally inspired by placing the containers to the nearest blocks provided that the operational capacity of the yard cranes of these block is not exceeded as constraint (10) dictates. The values of $\mathcal{R}^1$ are manipulated to give slightly different allocations from which newer (improved) allocations can be obtained.

Given a positive R_{qj}^{ti} value in $\mathcal{R}^1$, we could reallocate the containers from some block i to another block i'. This is done for each t, q, j and i. The values of R_q^{ti} are unaltered, as there is no point in re-positioning a container with an unknown future pick-up location.

To choose a block i' to reallocate containers to from i, we make use of random sampling where probability of choosing i' is inversely proportional to the driving time between i' and the future berth location j of the containers and is equal to:

$$\frac{(t_{i'j})^{-1}}{\sum_{i\in B}(t_{ij})^{-1}} \tag{20}$$

To determine the quantity we reallocate, we first have to determine the maximum amount of containers x that we can reallocate to block i' from i. Define $\chi = \lfloor \Delta \times (\mu_y - \epsilon) \rfloor$ as the maximum number of containers sent to a block per decision epoch — determined from (10). We have that $x = \min\{R_{qj}^{ti}, \chi - \sum_{q\in Q}(\sum_{j\in L} R_{qj}^{ti'} + L_q^{ti'})\}$ as the reallocated quantity is bounded by the number of containers block i' can receive in epoch t before its working capacity is exceeded. Given x, the probability that c containers are reallocated from i to i' is given by:

$$\frac{\chi - c}{\sum_{f=0}^{x}(\chi - f)} \tag{21}$$

This distribution is skewed against reallocating (many) containers as small values of c are more probable to preserve the feasibility in previous solutions with regards to the working capacities of the yard and quay cranes as most allocations are infeasible. Thereafter, we check whether the newly generated solutions satisfy constraint (19) before it can be added to the pool of solutions. The other constraints are met by default using this solution generation technique.

This procedure is repeated for $iter_1$ times with $\mathcal{R}^1$ being the original solution. This leads to the creation of a population of at most $iter_1$ solutions in the first iteration. Of these $iter_1$ solutions, the best pop solutions are picked and the rest are discarded. The solutions are ranked in terms of the objective value (6).

Now that we have a new population of solutions, we use each to generate at most $iter_1$ solutions using the approach described above with $\mathcal{R}^1$, and again we keep the best pop solutions overall. This procedure is repeated for $iter_2$ rounds, after which we return the best solution from the final population of pop solutions. The meta-heuristic is summarized in Algorithms 2 and 3 in Appendix B.

The heuristic largely resembles a single-parent genetic algorithm as multiple solutions are generated from one solution through a perturbation operation. Since we are dealing with a non-linear optimization problem (see Section 4.1), genetic algorithms render

themselves as a suitable approach. However, the 'localized' search in neighborhoods of our approach preserves the solution structure and, thus, the possibility to find feasible solutions in a heavily constrained problem. This in contrast to the randomized solution generation in genetic algorithms.

The parameters of our meta-heuristic were chosen by means of experimentation. A simple grid search may be applied to configure them. We noticed that the performance of the heuristic does not vary significantly with one specific parameter, so that it can be applied to different instances with ease. Rather, the choice of parameters should be focused on minimizing the computation time for real-time setting by limiting the total number of iterations — since its run-time complexity is in the order of $iter_2 \times iter_1 \times pop$.

With the above considered, our methodology is now almost completely described. The only remaining part is the provision of an approach by which we are able to estimate the quay crane productivity λ_q. The following section deals with this.

5. QC productivity estimate

In this, section we describe the procedure by which we arrive at estimates $\hat{\lambda}_q^t$ of the quay crane productivity at time (epoch) t. We first provide a theoretical analysis by which we are able to exactly determine the actual quay crane productivity λ^q under some restrictive assumptions in Section 5.1. In practice, some of the assumptions might not hold, and alongside the limited available computation time, we arrive at an approximation in Section 5.2.

5.1. Theoretical analysis

To exactly determine the quay crane productivity values, we establish a series of assumptions. First, we assume that the terminal system is in equilibrium so that the results concerning long-run time averages hold. This also dictates that trucks execute the same movements (cycles) throughout. We also assume that the service times of quay cranes follow an exponential distribution with rate μ_q. The choice of the exponential distribution in queuing system is quite popular in general as it simplifies the analysis. Furthermore, Guan and Liu (2009) mention that it could be very suitable for modeling waiting times in terminals. In addition to the uncertainty, we also assume that the service time for a discharge order follows the same distribution as that for a loading order. Analogous to the quay cranes, we assume that the yard crane service times follows an exponential distribution with rate μ_y. We also assume that service time distribution is identical for both grounding and retrieval.[4] In Section 5.2, we will analyze the situation when we deviate from these assumptions, of which some are relaxed our simulations in Section 6.

In the context of queuing theory, we treat the quay crane as a server instead of a customer – although it awaits service by trucks during discharge – as done in Liu and Ge (2018). Consider an arbitrary truck indefinitely rotating between a quay crane q and a block i. One can see that the cycle between two consecutive arrivals at q for a given truck, is identical to the cycle between two consecutive departures. Therefore, to measure the quay crane productivity, it suffices to look at the truck arrivals. Therefore, we consider the process renewal $\{N_q(t), t > 0\}$ counting the number of truck arrivals at q up to time t, with $N_q(0) = 0$.

Now suppose that there are V_i^q trucks serving crane q and block i indefinitely. Clearly, it holds that $n_q = \sum_{i \in B} V_i^q$. We define the following variables $N_{iq}^r(t)$ describing the number of times truck r arrived at quay crane q from block i in the interval $[0, t]$. Let k^r be the time that truck r first leaves q. This means that its renewal process at q could not have been initialized at time t for $t < k^r$. That said, we define its renewal process as follows $\{N_{iq}^r(t - k^r), t > 0\}$ and $N_{iq}^r(y) = 0$ for $y \leq 0$. The relationship between $N_q(t)$ and $N_{iq}^r(t - k^r)$ is given as follows:

$$N_q(t) = \sum_{i \in B} \sum_{r=1}^{V_i^q} N_{iq}^r(t - k^r) \tag{22}$$

Using results from renewal theory, we are interested in the arrival rate λ_q of trucks at q given by:

$$\lambda_q = \lim_{t \to \infty} \frac{N_q(t)}{t} = \lim_{t \to \infty} \frac{\sum_{i \in B} \sum_{r=1}^{V_i^q} N_{iq}^r(t - k^r)}{t} = \lim_{t \to \infty} \frac{\sum_{i \in B} \sum_{r=1}^{V_i^q} N_{iq}^r(t)}{t} \quad = \lim_{t \to \infty} \frac{\sum_{i \in B} V_i^q N_{iq}(t)}{t} = \sum_{i \in B} V_i^q \lim_{t \to \infty} \frac{N_{iq}(t)}{t}$$

where the second last equality only holds because of the assumption we made regarding the independence of truck movements, so that the delay of one truck does not affect other trucks traversing the same path. The validity of this assumption is assessed in Appendix H.1. The last equality implies that the long-run arrival rate from block i, is equal to the number of trucks V_i^q serving the path to block i from q multiplied by the long-run arrival rate at q of one of those trucks.

Continuing Eq. (23) and using that $\lim_{t \to \infty} \frac{N(t)}{t} = \lambda = \frac{1}{E(c)}$ for a general renewal process $\{N(t), t > 0\}$ with rate λ and expected cycle length $E(c)$, we have that:

$$\lambda_q = \sum_{i \in B} \frac{V_i^q}{E(e_q^i)} = \sum_{i \in B} \frac{V_i^q}{E[w_i] + t_{ij} + t_{ji} + E[w_q^i]} = \sum_{i \in B} \lambda_q^i \tag{23}$$

with $E(e_q^i)$ being the expected cycle length of a truck serving quay crane q and block i, $E[w_i]$ the expected waiting time of a truck at block i and $E[w_q^i]$ the expected waiting time at q for a truck serving block i. Note that the cycle length is equal to the sum of the travel times and waiting times between q and block i between two consecutive arrivals and that the waiting times are over-estimated as a truck, in reality, may depart from a yard crane before the container is grounded and the quay crane does not wait for a truck before it begins to process a discharge order. The distribution of e_q^i is unknown.

[4] In practice, retrieval operations may take longer as containers may have to be reshuffled. However, the complexity of the logistic processes involved make the relevance of this assumption minor, in addition to simplifying the analysis as a whole.

We still have to determine $E[w_q^i]$ and $E[w_i]$ in order to determine λ_q. However, their values depend on the arrival rate λ_q^i [5] of trucks serving block i at q which is what we are trying to calculate. The inter-dependency identified here compels us to develop a recursive approach by which the associated performance measures could be computed.

The standard approach in determining $E[w_q^i]$ - and $E[w_i]$ or its distribution is based on determining the steady-state distribution of the Discrete Time Markov Chain (DTMC) of the stochastic process describing the movement of trucks throughout the terminal. Specifically, Let $\{X_n, n \geq 0\}$ be this stochastic process. If we define the status X_n to be the number of trucks at each station in the terminal — where we also treat the travel times as stations with infinite servers and deterministic service time t_{ij} - after the nth truck arrival at q, the process is a DTMC. The choice of modeling the travel times as service stations is inspired by Choobineh et al. (2012).

In Tijms (2003), it is shown how a system of equations can be formulated to calculate the long-run equilibrium probability of observing a certain status $s \in S$, where S is the state space representing all possible truck allocations in the terminal between any crane $q \in Q$ and block $i \in B$. Practically, this approach is undesirable due to the computational burden associated with a large state space even for a relatively small numbers of stations (yard cranes, quay cranes and travel time stations) and trucks.

Despite the fact that $\{X_n, n \geq 0\}$ has a rather complex DTMC, the queuing network representing the terminal possesses a very desirable feature; that it is a BCMP network. BCMP networks were first introduced in Baskett et al. (1975). They are basically networks with specific server disciplines for which the long-run probability distribution of a state assumes a product form of the M/M/1 queue long-run distribution of its servers. The queuing system of our terminal is one such network, as per (Tijms, 2003), because the service discipline at each station is either characterized by FCFS with exponential service times (as with the cranes) or infinite service stations such that each customer (truck) gets assigned to a server immediately (as with the travel times).

Some interesting theorems hold for these networks, such as the arrival theorem, as mentioned in Tijms (2003). The arrival theorem implies that in the long-run, the number of customers (trucks in our case) seen by an arriving customer at each station **not counting that customer** is equal to the long-run number of customers at each station in an identical network without that customer. The implications of this theorem will be unfolded below.

We now present a mean-value algorithm that estimates $E[w_i]$, $E[w_q^i]$ and λ_q^i based on the arrival theorem, without having to calculate the probability of observing status s. Consider a single-truck scenario where the expected cycle length $E(e_q^i)$ is given by $E(e_q^i) = \frac{1}{\mu_y} + t_{ij} + t_{ji} + \frac{1}{\mu_q}$, from which we can automatically determine λ_q using $\lambda_q = \frac{1}{E(e_q)}$. Now that we have determined λ_q and $E[w_q]$, we may estimate the expected number of trucks L_q at the queue of q at any point in time by applying 'Little's Law'. The law is discussed in greater detail in Tijms (2003). In our context, this means that the expected number of trucks L_q at q is given by $L_q = \lambda_q E[w_q]$. The expected number of trucks L_i at yard block i can be determined in a similar manner, but with $E[w_i]$ instead of $E[w_q]$, while the arrival rate λ_i at block i would then equal to λ_q. The single truck scenario will be used as a termination criterion for our recursive algorithm.

Suppose we have multiple trucks. To continue the analysis, we introduce the allocation vector A. This is a vector with $|B|$ elements specifying the V_i^q for each $i \in B$. To emphasize the specification of the V_i^q values as per A, we use the notation $V_i^q(A)$. We also refer to $L_q(A)$, $E[w_q(A)]$ and $\lambda_q(A)$ as the relevant performance measures under allocation A. Similar notation holds for the performance measures of the yard crane queues. The arrival theorem implies the following:

$$E[w_q^i(A)] = \frac{1}{\mu_q}(1 + L_q(A - e[i])) \tag{24}$$

with e_i being a unit vector with ith element equal to 1 and all other elements equal to 0. (24) only holds when we have one server in the queue, that is namely q. The situation is more complex with multiple servers. A proposed approximation for this case can be found in Suri et al. (2007).

The implication of (24) is that it allows us to determine $E[w_q^i(A)]$ without knowing the arrival rate $\lambda_q^i(A)$. The latter can then be determined using that:

$$\lambda_q^i(A) = \frac{V_i^q(A)}{E[w_i(A)] + t_{ij} + t_{ji} + E[w_q^i(A)]} \tag{25}$$

In turn, the quantity $L_q^i(A)$, which represents the expected number of trucks can be determined using Little's Law; or $L_q^i(A) = \lambda_q^i(A)E[w_q^i(A)]$. The total arrival rate $\lambda_q(A)$ at q is equal to $\lambda_q(A) = \sum_{i \in B} \lambda_q^i(A)$ as given in (23). Likewise, the total number of trucks $L_q(A)$ at q under allocation A is equal to $sum_{i \in B} L_q^i(A)$. $L_q(A)$ can then again be used in (24) to calculate the waiting times at q under an allocation with one more truck. A similar analysis follows when determining $E[w_i(A)]$ and $L_i(A)$. Additionally, note that $\lambda_i = \lambda_q^i$.

The analysis naturally extends to the standard case with multiple quay cranes. In this setting, the vector A becomes a matrix with dimension $|Q| \times |B|$ with elements $V_i^q(A)$ for each $q \in Q$ and $i \in B$. Additionally, e_i would also become a binary matrix $e[q, i]$ with exactly one element equal to 1; element (q, i). Notation adjustment is also necessary, so that we have $E[w_i^q(A)]$ instead of $E[w_i(A)]$ as the waiting time of a truck at block i serving quay crane q. Analogously, we have $L_i^q(A)$ being the expected number of trucks from q at block i under allocation A. $L_i(A) = \sum_{q \in Q} L_i^q(A)$ is now the expected total number of trucks at block i and $\lambda_i(A) = \sum_{q \in Q} \lambda_q^i(A)$ is the expected total arrival rate at block i. The following set of equations define the relevant relationships:

$$E[w_q^i(A)] = \frac{1}{\mu_q}(1 + L_q(A - e[q, i])) \tag{26}$$

[5] The waiting time at a queue depends on the arrival rate at the queue.

$$E[w_i^q(A)] = \frac{1}{\mu_y}(1 + L_i(A - e[q,i])) \tag{27}$$

$$\lambda_q^i(A) = \frac{V_i^q(A)}{E[w_i^q(A)] + t_{ji} + t_{ij} + E[w_q^i(A)]} \tag{28}$$

$$L_q^i(A) = \lambda_q^i(A) E[w_q^i(A)] \tag{29}$$

$$L_i^q(A) = \lambda_q^i(A) E[w_i^q(A)] \tag{30}$$

Using these aforementioned equations, we can construct a mean-value algorithm that computes the equilibrium performance measures of the terminal under a particular allocation A. The algorithm is summarized in the pseudo-code of Algorithm 4. The time complexity of the algorithm is exponential in the number of trucks in the terminal. Specifically, it is $\mathcal{O}(V!)$ where $V = \sum_{q\in Q}\sum_{i\in B} V_i^q$. $V!$ is a large number, even for moderate values of V. This is due to the arrival theorem compelling us to try every permutation of the trucks under a given allocation to determine the performance measures. In principle, one could speed up the algorithm slightly by storing the results of encountered allocations to avoid recomputing them as we do. However, the order of magnitude of the number of computations remains more or less similar, making it unsuitable for real-time optimization where solutions need to be generated quickly.

With the above considered, several restrictive assumptions were introduced to calculate the quay crane productivity λ_q. These include the equilibrium setting and the exponential service times of the quay and yard cranes. Such assumptions may not hold in practice, and our method needs to cope accordingly. Moreover, the distribution of the cycle length e_q^i remained unknown and the mean-value algorithm used to determine λ_q had an impractical run-time. We treat these issues in the following section.

5.2. Approximate analysis

This section presents an approximation of the mean-value analysis by which we can overcome the associated obstacles. Firstly, we present two important assumptions. Accordingly, we develop a predictive model for the quay crane productivity which we can use in our decision-making problem.

5.2.1. Long-run average assumption

We simply assume that the equilibrium results hold over the course of epochs of length Δ. This is emphasized in (4) as the estimated arrival rates $\hat{\lambda}_q^t$ in epoch t are generated from data on long-run averages as will be explained in Section 5.2. This corresponds to a Pointwise Stationary Approximation (PSA). PSA is analyzed thoroughly in Green and Kolesar (1991) who conclude that it often serves an accurate approximation to the continuous-time case.

5.2.2. Cycle length distribution

The probability distribution of e_q^i defines the arrival process $\{N_q(t), t > 0\}$. This allows us to calculate the probability of meeting the target processing time during discharging with a certain probability. The calculation of this probability is necessary to determine the target productivity λ_q^* through Algorithm 1. Since we cannot determine the distribution of the cycle lengths, we have to propose one. Some factors ought to be considered first.

It is important that the distribution of e_q^i is such that the distribution of the merged renewal process in (22) is known. This places some restrictions on the possible distributions that we could use. For instance, if e_q^i is normally distributed with mean $\frac{V_i^q}{\lambda_q^i}$, then although the renewal process at (22) has rate λ_q, its inter-occurrence times – across all types i – are not normally distributed with mean $\frac{1}{\lambda_q}$. In fact, the inter-occurrence times follow an unknown distribution with no closed-form expression. Additionally, we seek a distribution such that the truck arrival process is identical to the truck departure process as explained in Section 5.1.

With the factors mentioned above in concern, one very suitable distribution is the exponential distribution for e_q^i. Firstly, if the inter-arrival times e_q^i of a truck follow an exponential distribution with rate $\frac{\lambda_q^i}{V_i^q}$, then the arrival process of the truck is Poisson with rate $\frac{\lambda_q^i}{V_i^q}$. In that case, the merged renewal process in (22) is Poisson with rate $\lambda_q = \sum_{i\in B}\lambda_q^i$. Moreover, the choice of exponential distribution results in a truck departure process that is identical to the arrival process; a Poisson with rate λ_q. This holds as a consequence of Burke's Output Theorem; see Burke (1956).[6] The validity of this choice will be assessed explicitly in Appendix H.2.

That said, the renewal process describing the total number of truck departures from crane q at time s $\{N_q(s), s > 0\}$ in (D.1) is a non-stationary Poisson process with rate $\lambda_q(s)$ at time s. In Appendix D, we explain how $P(\lambda_q^{[1,\dots,T]})$ can be calculated for this process. The waiting times at the blocks $\hat{w}_i^t$ could also be approximated by the mean waiting time in an $M/M/1/K$ queue — which defines $g(.)$ in Constraint (11), with $K = \sum_{q\in Q}(R_q^{ti} + \sum_{j\in L} R_{qj}^{ti} + L_q^{ti})$, the total number of container orders that block i ought to handle in epoch t. Practically, such a limit on the queue length may also exist as dictated by the number of available buffer zones near the block. This approximation is assessed in Appendix H.3.

[6] The theorem simply states that if the arrival process in a stationary queuing system is also Poisson with rate λ, then so is the departure process.

5.2.3. Arrival rate prediction

We need to devise a technique by which the parameters do not require extensive computation. To do that, we make use of machine learning where we try to predict the quay crane productivity λ_q using the number of trucks n_q trucks serving crane q, the average waiting times at the yard $\overline{w_q}$ of the trucks and the average driving time to and from the yard $\overline{t_q}$ of the trucks. We conjecture that these three measures are sufficient to determining the arrival rate λ_q. We will verify this below. Let $f(.)$ represent the regression model we use to make this prediction. Using mean value analysis, we construct a representative data set containing samples of λ_q and the corresponding values of n_q, $\overline{w_q}$ and $\overline{t_q}$.. This data is then used to train our machine learning model.

To generate the data, we pick an arbitrary crane q in the terminal. We sample the number of trucks n_q from an (integer) uniform distribution between υ_l and υ_u, with υ_l corresponding to the minimum number of trucks that crane could have, and υ_u being the maximum. We chose υ_l and υ_u to be 6 and 9 as these values characterize the scenarios we are interested in. We then pick a subset of n_{c1} trucks from the n_q trucks and assign it to some block i. We then pick another subset n_{c2} of the remaining $n_q - n_{c1}$ trucks and assign it another block $i' \neq i$. The procedure is repeated until each of the n_q trucks have been assigned to some block. No block is assigned more than one subset. A block is sampled randomly with equal probability among all blocks, while the size of each subset of trucks is uniformly sampled between 1 and $n_q - \sum_{r=1}^{R} n_{cr}$ with n_{cr} being the number of sampled trucks at the rth iteration and R being the total number of sampling iterations so far.

To better incorporate the effects of trucks from other quay cranes – not q – on the waiting times at the blocks, we introduce 'distortion trucks'. We are not interested in the rate by which they arrive at their quay cranes. Their function is to simply add traffic to the yard cranes that serve the trucks of q. The number of such trucks is sampled uniformly between 0 and $\upsilon_{max} - n_q$. The choice of υ_{max} should be such that Algorithm 4 is still capable to retain the performance measures efficiently; in the worst case, its running time is $\mathcal{O}(\upsilon_{max}!)$. We found $\upsilon_{max} = 15$ to be reasonable considering the equipment specification that we will define in our numerical experiments. For the distortion trucks, the above assignment procedure is used except that the blocks are chosen **only** from among those that the trucks of q serve, and a block may be chosen in multiple sampling iterations.

For the given $V_i^q(A)$ values, we can apply Algorithm 4 in order to compute the relevant performance measures for q. $\overline{t_q}$ and $\overline{w_q}$ could then be determined as follows:

$$\overline{t_q} = \frac{\sum_{i \in B} V_i^q(A)(t_{qi} + t_{iq})}{\sum_{i \in B} V_i^q(A)} \tag{31}$$

$$\overline{w_q} = \frac{\sum_{i \in B} V_i^q(A) E[w_i^q(A)]}{\sum_{i \in B} V_i^q(A)} \tag{32}$$

The advantage of averaging measures in (31) and (32) is that we do not deal with the assumption of fixed truck assignments anymore. Rather, we have that for given (mean) waiting and driving times, a corresponding arrival rate just like in our optimization model in Section 4.

With $n_q = \sum_{i \in B} V_i^q(A)$ already determined, we have a complete set of input features for building the model in (5). We generate a total of 2000 samples for our training data.

The next step is to choose a machine learning algorithm to forecast λ_q. We consider the following three regression algorithms: (1) Neural Network (NN), (2) K-Nearest Neighbors (KNN), and (3) Random Forests (RF). More information on these machine learning algorithms among others can be found in James et al. (2013).

Out of the three algorithms, we choose the one with highest forecast accuracy. In order to assess the accuracy of the algorithms, we use nested cross validation. Details on the nested cross-validation and the grid of parameters considered can be found in Appendix E. The results of the nested cross-validation are reported in Table 1. We observe that all algorithms have an R^2 of at least 0.95, implying that a model that predicts the arrival rates based on only the three regressors n_q, $\overline{t_q}$ and $\overline{w_q}$ produces forecasts of reasonable quality.

Table 1
Outer cross-validation R^2 scores – in percentages – for each of the algorithms considered in the nested cross-validation.

Algorithm	Performance	
	Mean (%)	Standard deviation (%)
NN	96.04	±0.48
KNN	95.48	±0.80
RF	95.41	±0.87

From these results, we deduced that the NN with one hidden layer containing a 100 neurons is the best algorithm as it has a higher mean R^2 and a lower standard deviation. This is likely because of the NN's ability to capture complex relationships between variables which other simpler algorithms might not (James et al., 2013). It is worthwhile mentioning that the NN is, however, slightly slower in producing forecasts. Therefore, if the machine learning algorithm is called many times during the optimization, it might be wiser to use one of the other two algorithms, given their speed in generating predictions and acceptable accuracy.

As explained above, normal cross-validation is again applied to choose the best parameters of the NN out of those reported in Table E.6.[7] The resulting NN is then trained on the full data set with 2000 samples to give our $\hat{f}(.)$. Although, the NN is trained

[7] The chosen NN had a 'logistic' activation function, an 'lbfgs' solver and an 'adaptive' learning rate.

from a dataset with restrictive assumption such as exponential crane service times, it can still be a useful tool in an optimzation setting. In Appendix H.4, we examine the practicality of using the forecasts obtained from $\hat{f}(.)$ in a real-time setting where some of the assumptions in Section 3 do not hold anymore.

One must not forget however that the $\hat{\lambda}_q^t$ values estimated by $f(.)$ are under-estimates of the actual arrival rates. This is due to the over-estimations of the truck waiting times at the yard and the quay in Algorithm 4 - which is used to generate the data for training $\hat{f}(.)$ - by assuming that truck service times exactly coincide with the containers'.

6. Numerical experiments

This section deals with the numerical experiments. We first discuss the experimental set up, and then introduce a set of scenarios under which we will compare the allocation policies by means of a simulation. A detailed discussion of the results and their implications follows after.

6.1. Set up

This section discusses the parameter construction. Since we have no access to real-life data, we resorted to generating our own data. Not to mention that real-life data offers no guarantee to how our method generalizes to other terminals. We also describe the benchmark policy with which we compare our approach.

For the generation of the vessel schedule, we assume a Poisson arrival process as observed in several studies such as El-Naggar (2010) and Lang and Veenstra (2010). A major advantage of using a Poisson process is robustness since Poisson allows high and low quay occupancy, due to small and large vessel inter-arrival times. By developing an allocation policy that is tested under vessel schedules witnessing variations in inter-arrival times, we can better guarantee that our method produces consistent results. Note that our method is invariant of the process by which vessels arrive, and the Poisson assumption is, thus, non-essential. The Poisson arrival rate is taken to be Γ, and describes the operational density of the terminal with higher values of Γ implying more frequent vessel arrivals. We generate vessel arrivals over a course of six weeks.

Since we only focus on the seaside, we decide on the total number of vessel-to-vessel containers C in the six week period beforehand. It is advisable for C to be proportional to both Γ in order to preserve the number of containers handled per vessel. For each of the C containers, we assume a discharging vessel and a loading vessel. The discharging vessel is chosen from a uniform distribution over all vessel arrivals during the six week period. We refer to the vessel by its arrival number, i.e. vessel x is the xth arrival. The loading vessel is also sampled uniformly in the interval $[x+\xi, x+\xi_{max}]$, where ξ and ξ_{max} are positive integers. Like C, it is advisable to let ξ vary with Γ to avoid situations when the loading vessel arrives while the discharging vessel is still at berth. ξ_{max} on the other hand can be fixed, and we set it to 50. If the loading vessel corresponds to an arrival number after the last arrival in the six week period, we assume that the container has an unknown pick-up location.

To decide on berth locations, we sample uniformly from the set $L^* \subset L$ of available locations. We chose that $|L| = 10$. Some locations may be unavailable as they are occupied by other vessels. One should be careful that $L^* \neq \emptyset$. We assign a vessel to an arbitrary location $j \in L^*$. We also decide on the number of quay cranes $|Q_b|$ operating on the vessel using a uniform distribution in the interval $[2, M]$ where M is the maximum number of allowed cranes per vessel. Like C, M may vary with Γ to account for larger numbers of containers needing service. The containers allocated to the vessel are then randomly allocated with equal probability among the $|Q_b|$ cranes.

To determine the berth times, we determine a certain level of quay crane productivity during both discharge and loading. The productivity level represents the crane utilization and may be different for discharge and loading, especially since loading involves longer waiting for trucks and less crane utilization. For discharge, we pick a utilization of $\mu_q - 0.025$ and for loading $\mu_q - 0.1$. The total time $\mathbf{t}^b$ that ship may spend at berth is thus $\mathbf{t}^b = \max_{q \in Q_b}\{C_q^d \times \frac{1}{\mu_q - 0.025} + C_q^l \times \frac{1}{\mu_q - 0.1}\}$ where C_q^d and C_q^l are the numbers of containers quay crane q has in its discharge and load work queues.

The total number of epochs $\mathbf{t}^b_{[N]}$ is thus equal to $\mathbf{t}^b_{[N]} = \left\lceil \frac{\mathbf{t}^b}{\Delta} \right\rceil$. For the target discharge time $\mathbf{t}^d_q$ per crane, the procedure is more involved. We firstly determine the total time a crane needs to finish its overall work queue — if it were to maintain the target productivity during loading and discharge. For crane q, the total processing time $\mathbf{t}^b_q$ is given by $\mathbf{t}^b_q = \{C_q^d \times \frac{1}{\mu_q - 0.025} + C_q^l \times \frac{1}{\mu_q - 0.1}\}$, and the corresponding number of epochs is given by $\mathbf{t}^b_{q[N]} = \left\lceil \frac{\mathbf{t}^b_q}{\Delta} \right\rceil$. The difference between $\mathbf{t}^b_{[N]}$ and $\mathbf{t}^b_{q[N]}$ represents an excess number of epochs that could be used by crane q in discharge time. The minimum time $\mathbf{t}^d_{q-\min}$ that crane q could spend on discharge is $\mathbf{t}^{\mathbf{d}}_{\mathbf{q}-\min} = \{C_q^d \times \frac{1}{\mu_q - 0.025}\}$, resulting in a minimum number $\mathbf{t}^d_{q-\min[N]}$ of discharge epochs: $\mathbf{t}^d_{q-\min[N]} = \left\lceil \frac{\mathbf{t}^{\mathbf{d}}_{\mathbf{q}-\min}}{\Delta} \right\rceil$. In turn the total number of epochs $\mathbf{t}^q_{d[N]}$ that q could spend on discharge is equal to $\mathbf{t}^q_{d[N]} = \mathbf{t}^d_{q-\min[N]} + \mathbf{t}^b_{[N]} - \mathbf{t}^b_{q[N]}$. The target discharge time $\mathbf{t}^q_d$ can then be determined from $\mathbf{t}^q_d = \mathbf{t}^q_{d[N]} \times \Delta$. Additionally, we reserve the berth location for an additional hour to ensure enough time for the vessel to depart the terminal before another vessel could berth in the same location.

Thereafter, we can determine the N^t_{qj}/N^t_q and L^{ti}_q parameters spreading the containers evenly among the berth's epochs. This does not mean however that a container cannot be discharged or loaded before the corresponding epoch t. In real-life cases, the quay crane is always in operation as long as there are is a truck awaiting service, and cannot be deliberately delayed. We account for this in our simulation. Such deviations present a disadvantage of the discrete-time model in Section 4.1. Additionally, one has to be careful when determining the L^{ti}_q parameters so that constraint (10) is not violated. For instance, if no more loading orders

for a quay crane could be issued to block i in epoch t, we loading orders to block $\hat{i} \neq i$ instead.

Lastly, we decide on the dimensions of the terminal that define the driving times, the number of blocks and berth locations. We have a quay extending to 3 km, fitting 10 berth locations; i.e $|L| = 10$ as each berth location extends to 300 m. We assume a yard that is parallel in length to the quay and fits 30 blocks as one block spans 100 m in width. The traveling times (in minutes) are symmetric in our case so $t_{ij} = t_{ji}$ and are given by the L_1 distance (in meters) between the midpoint of the berth location and the midpoint of a block divided by a constant speed of 333.3 meters per minute. This gives a minimum and maximum travel time of 0.6 and 9 min.

For the simulation model, we assume a normal distribution for the crane service times. This distribution is more in line with that encountered in practice as we learned from experts. The crane service times have means $\frac{1}{\mu_q}$ and $\frac{1}{\mu_y}$ and variances 0.25 and 1. Note that the yard crane's service time has a higher variance due to the longer distances it may travel along the rows and re-shuffling moves. We also truncate both service times at 1 min for the quay crane and 1.25 min for the yard crane to prevent unreasonably low values.

The benchmark policy is inspired by the disjoint optimization presented in Zhang et al. (2003). They partition the problem into two stages whereby they determine the work queues of the yard cranes in the first stage, and the container assignments in the second stage with no real-time quantification of the effect of allocation decisions on QC productivity. Their formulation is adjusted slightly to accommodate our setting and is given in Appendix G. To compare with our method, we consider a series of statistics on the percentage decrease in berth time. Among these statistics are confidence intervals on the mean berth time to account for variations in call size and yard occupancy. The usage of confidence intervals in berth-related studies has been used in works like (Sheikholeslami and Ilati, 2018).

6.2. Results

The policies are tested using a real-time simulation that takes as input our allocation decision for each container over the six-week period. In Appendix F, we explain the simulation model in greater detail, and provide a flowchart describing the sequence of events that take place.

In all scenarios, we choose the length of a decision epoch Δ to be 15 min and the number of epochs in a receding horizon $T = 10$, meaning that our planning horizon extends to 2.5 h. Additionally, the parameter values of the meta-heuristic in Section 4.2 have been determined through experimentation to be $iter_2 = 5$, $iter_1 = 3$, $pop = 10$ and $d = 0.05$. We construct an initial solution $\mathcal{R}^1$ by assuming a maximum capacity per block in each epoch — which we take to be 3 - and allocate the containers from a vessel to the closest blocks such that this capacity is not exceeded. In case the maximum capacity is insufficient, we increase the capacity by 1 for all blocks. When generating $\mathcal{R}^1$, we randomly select a container from all discharging quay cranes in all berth locations. The initial solution affects the final solution significantly and should be chosen with care.

In total, we propose three scenarios, each characterized by a certain level of operational density. The first scenario is characterized by parameters $\Gamma = \frac{1}{600}$, $C = 25000$ giving a total of $|\mathcal{B}| = 105$ vessels, $\xi = 3$ and $M = 3$, the second by $\Gamma = \frac{1}{480}$, $C = 30000$ giving a total of $|\mathcal{B}| = 128$ vessels, $\xi = 5$, $M = 3$ and the third scenario characterized by $\Gamma = \frac{1}{360}$, $C = 40000$, $\xi = 7$, $M = 4$ giving a total of $|\mathcal{B}| = 179$ vessels over the six week period. A comparison among the vessel schedules in each scenario can be found in Figs. J.18–J.20 in Appendix J, which display the epoch number on the x-axis and the berth location on the y-axis. The black points indicate an occupancy of the corresponding berth location and epoch.

For each scenario, there are two features that we want to test. Firstly, the standard of efficiency characterized by α. Higher demand for efficiency characterized by larger values of α require faster discharge processes. However, this also imposes constraints on the driving distances, which could prevent us from placing containers closer to future berth locations, thereby, slowing loading.

Another factor of interest is the number of trucks. Larger truck numbers often translate to larger arrival rates. We seek to investigate whether larger truck numbers contribute significantly to reductions in berth times, or are constrained by the quay crane's working capacity?

The results are presented in Tables 2–4. Here, 'Mean Decrease' refers to the mean percentage decrease in berth time over all vessel arrivals in the six week horizon by using our policy compared to Zhang et al. (2003). 'Economic Gain' explicitly states the expected annual cost saving with respect to Zhang et al. (2003). This KPI assumes that each extra minute of berth incurs some economic cost and compares the average berth times to see what is the average reduction in economic cost. It is an important assessment criterion from a managerial standpoint. It is simply the expected number of berths per year multiplied by the expected cost-saving per berth and is calculated as $1440 \times \Gamma \times (\psi_z - \psi_a) \times \Theta \times 365$, with ψ_z and ψ_a being the mean berth times under the policy of Zhang et al. (2003) and our proposed policy, and Θ is some penalty factor representing the marginal cost of an extra minute of berth per vessel. In principle, we have no data on typical values of Θ, but experts state that the order of magnitude of such a penalty can go up to tens of thousands of Euros per hour. Hence, we set Θ to $\frac{10{,}000}{60} \approx 167$ Euros per minute.

A typical pattern that is observed when applying our meta-heuristic for the $iter_2 = 5$ iterations when solving (6)–(17) is given in Fig. J.21 in Appendix J. The figure depicts the lowest attained value of (6) over the course of the iterations. A monotonic decrease is observed, although the steepness of the decline decreases over the course of the iterations as it becomes more difficult to drive further away to place the containers closer to their loading locations without significant delays.

From a computational perspective, a problem corresponding to a receding horizon is solved within a few seconds under both approaches, although the approach of Zhang et al. (2003) is faster. The difference becomes more prominent the more receding horizons we consider. Given an Intel(R) Core(TM) i5-6600K processor with 3.50 GHz and 16 GB RAM, the approach of Zhang et al.

Table 2
Results of Scenario 1, characterized by $\Gamma = \frac{1}{600}$, $C = 25000$, $\xi = 3$ and $M = 3$ with 105 vessel arrivals over a period of six weeks.

α	n_q	Mean Decrease	St. Dev	0.95 C.I.	Min.	Max	Economic Gain (€/year)
0.50	7	2.05%	3.64%	[1.34,2.75]%	−6.32%	10.94%	1,253,722
	8	1.43%	2.73%	[0.90,1.96]%	−6.41%	8.21%	721,220
	9	0.74%	2.99%	[0.16,1.32]%	−8.69%	12.88%	418,395
0.75	7	1.95%	3.76%	[1.22,2.67]%	−6.54%	11.47%	1,199,594
	8	1.18%	6.32%	[−0.05,2.4]%	−44.40%	34.43%	699,278
	9	0.49%	6.59%	[−0.78,1.77]%	−47.68%	34.96%	374,508

Table 3
Results of Scenario 2, characterized by $\Gamma = \frac{1}{480}$, $C = 30000$, $\xi = 5$, $M = 3$ with 128 vessel arrivals over a period of six weeks.

α	n_q	Mean Decrease	St. Dev	0.95 C.I.	Min.	Max	Economic Gain (€/year)
0.50	7	2.24%	5.72%	[1.24,3.24]%	−31.54%	26.12%	1,693,330
	8	1.12%	2.70%	[0.65,1.59]%	−7.50%	6.74%	852,151
	9	0.38%	5.63%	[−0.60,1.37]%	−35.87%	25.99%	462,648
0.75	7	1.95%	5.66%	[0.96,2.94]%	−32.24%	26.81%	1,521,437
	8	0.83%	2.59%	[0.37,1.28]%	−7.47%	6.92%	638,199
	9	0.41%	4.87%	[−0.44,1.26]%	−34.49%	27.29%	462,648

Table 4
Results of Scenario 3, characterized by $\Gamma = \frac{1}{360}$, $C = 40000$, $\xi = 7$, $M = 4$ with 179 vessel arrivals over a period of six weeks.

α	n_q	Mean Decrease	St. Dev	0.95 C.I.	Min.	Max	Economic Gain (€/year)
0.50	7	0.85%	14.23%	[−1.25,2.95]%	−96.17%	50.81%	1,501,931
	8	**−0.08%**	13.22%	[−2.03,1.87]%	−97.55%	50.11%	597,359
	9	**−0.16%**	11.71%	[−1.89,1.57]%	−99.36%	47.82%	268,202
0.75	7	1.15%	11.85%	[−0.60,2.9%]	−100.59%	51.50%	1,467,796
	8	0.33%	11.03%	[−1.30,1.95]%	−95.90%	48.88%	755,842
	9	**−0.04%**	11.81%	[−1.79,1.70]%	−95.22%	49.19%	409,618

(2003) takes less than five minutes to solve for all the receding horizons in the six week period using a Gurobi 10.0.3 solver, while our approach takes about 45 min. Both approaches were implemented in a Python 3.11 environment. In practice, however, one problem has to be solved every $\Delta \times T = 150$ min, so a difference of few seconds is likely to be trivial, as opposed to the difference encountered when solving for a full 6 weeks.

The results imply that our approach more often outperforms that of Zhang et al. (2003) than not. For most cases, the berth time decreases on average, going up to 2% for some cases. This is considered significant if such a reduction could only be attributed to tactical decisions, without incurring additional capital expenditures.

We also report the standard deviation, a 95% confidence interval for the mean percentage decrease, the minimum and maximum encountered values. The percentage decreases seems most variable for the 3rd scenario where berth inter-arrivals times vary most — due to higher arrival rate in Poisson process. The confidence intervals are often skewed to the positive side, and do not exclude a zero mean for half the cases. For scenario 3 specifically, the intervals is skewed to the negative side for cases with larger truck numbers, possibly due to yard congestion where some of our approximations may fall short. Lastly, the minimum and maximum percentage indicate dramatic reductions and elevations in berth time compared to Zhang et al. (2003) for nearly all scenarios, with the minimum mostly larger than the maximum.

Additionally, positive economic gains are encountered for all cases. Our method results in expected annual costs savings of several hundred thousand – and sometimes over a million – euros per year compared to Zhang et al. (2003). Such figures could amount to annual labor costs of several employees at the terminal or even purchase of new equipment.

In some cases such as in the third scenario, the mean decrease assumes smaller and sometimes negative values. Here, the margin for improvement is quite tight due to the elevated operational density with more than 10 quay cranes often discharging simultaneously.

Observe the disparity between the KPIs. While the mean percentage decrease in berth time is not always positive and assumes very small values for some cases, the corresponding economic gain which is based on the decrease in mean berth time overall is far from negligible. This calls for further analysis of the mean percentage decrease which we conduct in Appendix C. The conclusion of the analysis is that the approach of Zhang et al. (2003), while performing worse generally, may be better at preventing exceptionally long berth time during peak hours such as in Scenario 3.

A potential reason so as to why our approach often outperforms (Zhang et al., 2003) is that the disjoint optimization results in trucks driving too far. It is sometimes advantageous to wait longer by a close block as opposed to driving longer times to a less

occupied block. The disjoint optimization neglects this by spreading the work queues evenly among the yard cranes forcing some trucks to drive significantly longer distances. We do note, however, that the truck waiting times at the yard cranes are much less under the disjoint optimization.

Another reason for the reduction in berth time is that the loading locations of containers are disregarded under the policy of Zhang et al. (2003) which can significantly disrupt the loading operations if containers are placed further away.

On the other hand, our approach may be hindered by the variety of approximations involved. While our approximations often lay within a reasonable margin as shown in Appendix H, they sometimes deviate significantly which can lead to a poor estimate of the QC productivity corresponding to an allocation decision.

Furthermore, the mean percentage decrease in berth time decreases monotonically with the number of trucks. This is because the availability of more trucks leads to the quay crane operating at near maximum productivity regardless of the block-allocation policy.

With regards to α, and as we expected, it appears to have a negative effect when yard occupancy is manageable as seen in Scenarios 1 and 2. This could be explained by the restrictions imposed on the truck movements during discharge which disallow the containers from approaching their future berth locations leading to longer loading times. Such restrictions may however be advantageous with a high OYD as can be seen in Scenario 3 where $\alpha = 0.75$ leads to a larger mean percentage decrease. This could be for example due to the prevention of longer driving times when the waiting times at the yard are already high due to yard occupancy. In Appendix I, we motivate our probabilistic model by showing that the restrictions imposed by α are valuable compared to a setting where constraint (19) is completely ignored.

7. Conclusions

In this paper, we examined the potential of optimizing transshipment operations in real-time with the objective of minimizing vessel berth times. The decision involved here is the block-assignment of the containers. We considered a setting with vessel-to-vessel containers only, although the analysis presented can easily incorporate other carriers. Additionally, we assumed no knowledge of future vessel berth times as is the case in previous literature.

We presented a model that respects the trade-off between both discharge and loading operations by striving to place the containers as close as possible to their loading locations without delaying discharge significantly. To do that, we quantify effect of our block-allocation decisions on the quay crane productivity during discharge using the framework of renewal theory, queuing theory and machine learning. Thereafter, we formulated an optimization problem to decide on the block-allocation, where the objective minimizes the driving time during loading and constraints are imposed on performance measures during discharge.

The optimization problem is highly non-linear. Consequently, we proposed a meta-heuristic to solve it. We tested our method using a real-time simulation under different scenarios. Our results show that our method is more capable of minimizing vessel berth times under the scenarios considered compared to other established approaches such as Zhang et al. (2003) while introduces significant economic gains. The margin of improvement, however, decreases as the number of trucks increases due to the limit on the quay crane productivity. On the other hand, this also means that our approach is extremely reliable in cases where there is a shortage of trucks.

Our model is based on a set of assumptions which we impose for ease of analysis. Future research can be dedicated to estimating the performance measures more accurately when some of these assumptions do not hold. We are certain that more accurate estimations are capable of improving our methodology even further.

CRediT authorship contribution statement

Abdo Abouelrous: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Rommert Dekker:** Writing – review & editing, Supervision, Conceptualization. **Laurens Bliek:** Writing – review & editing, Supervision. **Yingqian Zhang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Abdo Abouelrous reports financial support was provided by European Supply Chain Forum.

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Appendix A. Table of parameters

See Table A.5.

Table A.5
Explanation of notation used.

Notation	Meaning
P_b	Probability that berth b meets its target discharge time.
L	Set of berth locations.
Q_b	Set of quay cranes $q \in Q$ at berth b.
B	Set of yard blocks.
T	Number of time periods/epochs in planning horizon.
C_q	Number of containers to be discharged by crane q.
μ_q	Maximum service rate of quay cranes.
μ_y	Maximum service rate of yard cranes.
λ_q	Long-run service rate/productivity of quay crane $q \in Q$.
λ_q^*	Target service rate for quay crane $q \in Q$.
λ_q^t	Service rate of quay crane $q \in Q$ in period t for $1 \le t \le T$. Its estimate is given by $\hat{\lambda}_q^t$
λ^q	Long-run service rate of quay crane $q \in Q$.
λ_i^t	Service rate of yard crane $i \in B$ in period t for $1 \le t \le T$. Its estimate is given by $\hat{\lambda}_i^t$
e_q^i	Cycle length of a truck driving between quay crane $q \in Q$ and yard block $i \in B$.
t_{ji}	Driving time from berth location $j \in L$ to block location $i \in B$. Similar definition holds for t_{ij}.
w_q^t	Expected waiting time at crane $q \in Q$ in period t for $1 \le t \le T$.
w_i^t	Expected waiting time at block $i \in B$ in period t for $1 \le t \le T$. Its estimate is given by $\hat{w}_i^t$
n_q	Number of trucks serving crane $q \in Q$.
$\overline{t_q^t}$	Nean driving time for a truck serving crane $q \in Q$ in period t for $1 \le t \le T$.
$\overline{w_q^t}$	Nean waiting time at the yard for a truck serving crane $q \in Q$ in period t for $1 \le t \le T$.
N_{qj}^t	Number of containers discharged by crane $q \in Q$ in period t for $1 \le t \le T$ to be picked up at berth location $j \in L$.
N_q^t	Number of containers discharged by crane $q \in Q$ in period t for $1 \le t \le T$ with unknown pick-up location or picked up at the land-side.
R_{qj}^{ti}	N_{qj}^t containers allocated to block $i \in B$.
R_q^{ti}	N_q^t containers allocated to block $i \in B$.

Appendix B. Algorithms

Algorithm 1 Bisection Method(α)

Input: Target probability α
Output: Arrival rate c that meets α
Initialization: Lower bound a, upper bound b, tolerance parameter ϵ
$c \leftarrow \frac{a+b}{2}$
while $\frac{b-a}{2} > \epsilon$ **do**
  if $P_q(c) - \alpha = 0$ **then**
    return c
  else if $(P_q(a) - \alpha) \times (P_q(c) - \alpha) < 0$ **then**
    $b = c$
  else
    $a = c$
  end if
  $c = \frac{(a+b)}{2}$
end while
return c

Algorithm 2 Block Allocating Algorithm(N,L)

Input: Population size pop, Discharge orders N, Loading orders L during planning horizon of T epochs.
Output: Allocation of discharge containers to blocks $\mathcal{R}^*$.
Initialization: Generate initial solution $\mathcal{R}^1$ using some rule of thumb.
Let $\mathcal{S}$ be the current solution population, $\mathcal{S} = \{\mathcal{R}^1\}$.
for $i2$ in $iter_2$ **do**
  $\overline{\mathcal{S}} = \mathcal{S}$
  for $solution$ in $\mathcal{S}$ **do**
    for $i1$ in $iter_1$ **do**
      alt_sol =Create Alternative Solution($solution$)
      if alt_sol is feasible **then**
        $\overline{\mathcal{S}} = \overline{\mathcal{S}} \cup \{alt_sol\}$
      end if
    end for
  end for
  $\mathcal{S} = pop$ solutions with lowest value for (6) in $\overline{\mathcal{S}}$.
end for
R^* = solution in $\mathcal{S}$ with lowest value for (6).
return R^*

Algorithm 3 Create Alternative Solution(R)

Input: Proposed allocation of containers R.
Output: Alternative allocation of container $\overline{R}$.
Initialization: $\overline{R} = R$
for t in T, i in B, q in Q, j in J **do**
  if $R^{ti}_{qj} > 0$ **then**
    Sample i' according to (20) for given j.
    $x = \min\{R^{ti}_{qj}, \chi - \sum_{q\in Q}(\sum_{j\in L} \overline{R^{ti'}_{qj}} + L^{ti'}_{q})\}$
    Sample $quantity$ according to (21) for given x.
    $\overline{R}^{ti'}_{qj} = \overline{R}^{ti'}_{qj} + quantity$
    $\overline{R}^{ti}_{qj} = \overline{R}^{ti}_{qj} - quantity$
  end if
end for
return $\overline{R}$

Algorithm 4 Terminal Performance Measures(A)

Input: Allocation A of trucks among blocks for each quay crane.
Output: Matrices $\lambda_Q^I(A)$, $E[w_Q^I(A)]$ and $E[w_I^Q(A)]$ containing $\lambda_q^i(A)$, $E[w_q^i(A)]$ and $E[w_i^q(A)]$ values $\forall q \in Q, i \in B$ under allocation A.
Initialization: $\lambda_q^i(A) = E[w_q^i(A)] = E[w_i^q(A)] = 0 \ \forall q \in Q, i \in B$.
if A is an allocation for a single truck scenario **then**
    $q \leftarrow$ the crane which the truck serves
    $i \leftarrow$ the block which the truck serve.
    $E[w_q^i(A)] = \frac{1}{\mu_q}$, $E[w_i^q(A)] = \frac{1}{\mu_y}$. Use (28) to determine $\lambda_q^i(A)$
    return $\lambda_Q^I(A)$, $E[w_Q^I(A)]$, $E[w_I^Q(A)]$
end if
for $q \in Q$ **do**
    for $i \in B$ **do**
        if $V_i^q > 0$ **then**
            $\lambda_Q^I(A - e[q,i])$, $E[w_Q^I(A - e[q,i])]$, $E[w_I^Q(A - e[q,i])]$=Terminal Performance Measures$(A - e[q,i])$
            Determine $L_i^q(A - e[q,i])$ using (30)
            Determine $L_q^i(A - e[q,i])$ using (29)
            $L_q(A - e[q,i]) = \sum_{i \in B} L_q^i(A - e[q,i])$
            $L_i(A - e[q,i]) = \sum_{q \in Q} L_i^q(A - e[q,i])$
            Determine $E[w_q^i(A)]$ using (26).
            Determine $E[w_i^q(A)]$ using (27).
            Determine $\lambda_q^i(A)$ using (28)
        end if
    end for
end for
return $\lambda_Q^I(A)$, $E[w_Q^I(A)]$, $E[w_I^Q(A)]$

Appendix C. Analysis of mean percentage decrease in berth time

In this section we discuss the distribution of the percentage decrease in berth time per vessel in the 6 week schedule for each of the 3 scenarios. We refer to Figs. C.3–C.5 that show the histograms for the distribution of the percentage decrease in berth time e for Scenarios 1, 2 and 3. The figures concern all berths throughout the 6 weeks and are shown for the case $n_q = 7$ and $\alpha = 0.5$.

We can observe that the distributions are skewed to the positive side, implying that decreases in berth time are more frequent than increases. Most values lay in the interval of 0%–5%. However, there are some very distant outliers, especially in Scenarios 2 and 3. The negative outliers are considerably large in magnitude which explain the low averages observed in Section 6. There are also some positive outliers that partially counter the effect of the negative ones, although they are smaller in absolute terms.

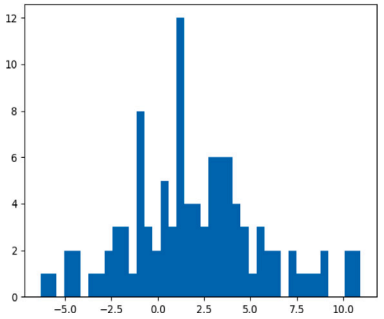


Fig. C.3. Distribution of percentage decrease in berth time for Scenario 1 with $n_q = 7$ and $\alpha = 0.5$.

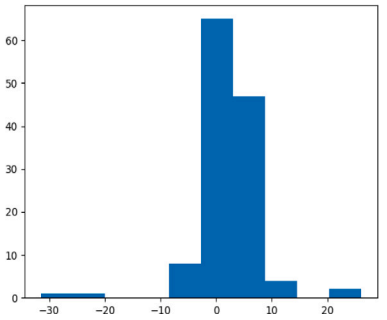


Fig. C.4. Distribution of percentage decrease in berth time for Scenario 2 with $n_q = 7$ and $\alpha = 0.5$.

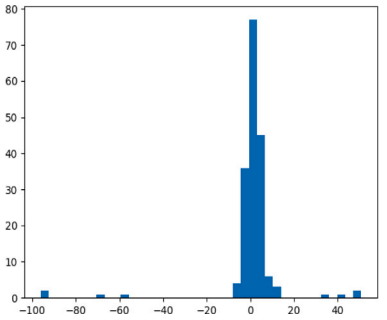


Fig. C.5. Distribution of percentage decrease in berth time for Scenario 3 with $n_q = 7$ and $\alpha = 0.5$.

From our in depth analysis of the results, such negative outliers occur during peak ours when the Quay has a large throughput and the operational yard density is high. The associated traffic leads to bottlenecks at the yard which can cause significant delays if not properly managed.

The approach of Zhang et al. (2003) is better at managing such peaks due to the way it spreads the workload among the yard cranes. It is aided by the ample amount of yard cranes (30) in our terminal. While managing such peaks is important for terminals from an operational perspective, they occur rarely as vessel planning is often done in such a way to prevent them.

Appendix D. QC productivity approximation

Let $N_q(s)$ be a random variable describing the number of trucks that arrived at q up to time s. Relating this to (2), we have the following:

$$P_b = \Pi_{q \in Q^b} P(N_q(\mathbf{t}_q^d) - N_q(\mathbf{t}_q^a) \geq C_q) = \Pi_{q \in Q^b} (1 - \sum_{n=0}^{C_q - 1} P(N_q(\mathbf{t}_q^d) - N_q(\mathbf{t}_q^a) = n)) \tag{D.1}$$

The rightmost expression can be calculated depending on the associated renewal process. Since we assume an exponential distribution for the truck inter-arrival times — as described in Section 5.2.2, the renewal process $N_q(s)$ is a non-stationary Poisson process with arrival rates $(\lambda_q^{[1,\dots,T]})$. In other words, we approximate $P(N_q(\mathbf{t}_q^d - \mathbf{t}_q^a) \geq C_q)$ by $P(\lambda_q^{[1,\dots,T]})$ under a non-stationary Poisson Process. For the total number of truck departures $N_q(s)$ at time s, the probability of observing n departures in a time interval $[s, s+x]$ - for positive x - under a non-stationary Poisson Process is given by:

$$P(N_q(s+x) - N_q(s) = n) = e^{-[M_q(s+x) - M_q(s)]} \frac{[M_q(s+x) - M_q(s)]^n}{n!} \tag{D.2}$$

with the mean-value function $M_q(s)$ satisfying:

$$M_q(s) = \int_0^s \lambda_q(z) dz = \sum_{t=1}^{T-1} \int_{\Delta(t-1)}^{\Delta t} \lambda_q^t dz + \int_{(T-1)\Delta}^{s} \lambda_q^T dz = \sum_{t=1}^{T-1} \Delta \lambda_q^t + [s - \Delta(T-1)] \lambda_q^T \tag{D.3}$$

for $T = \left\lceil \frac{s}{\Delta} \right\rceil$ and Δ being the length of the planning epoch t during which the rate λ_q^t is assumed to be constant. In turn, $P(N_q(\mathbf{t}_q^d - \mathbf{t}_q^a) \geq C_q)$ is equal to:

$$= 1 - \sum_{n=0}^{C_q - 1} e^{-[\Delta \sum_{t=t_1^q}^{t_T^q} \lambda_q^t]} \frac{[\Delta \sum_{t=t_1^q}^{t_T^q} \lambda_q^t]^n}{n!} \tag{D.4}$$

Appendix E. Nested cross validation

Nested cross-validation is explained more precisely in Cawley and Talbot (2010). The major idea behind it is that cross-validation is carried out on two levels to prevent a bias in evaluating the performance of algorithms when making a choice on their parameters. In the first (outer) level, the data is split between training and a testing set to constitute the outer cross-validation sets. In the second (inner) level, the training set is split into a sub-training and sub-testing set that constitute the inner-cross validation's data. The inner cross-validation chooses the best parameters according to some criterion. On the other hand, the outer cross-validation determines the forecast quality of the algorithm using some criterion. Our assessment criteria for both the inner and outer cross-validation is the Explained Variance Score, also known as the R^2. It is given by:

$$R^2 = 1 - \frac{\sum_{n=1}^{N}(\lambda_n - \hat{\lambda}_n)^2}{\sum_{n=1}^{N}(\lambda_n - \overline{\lambda})^2} \tag{E.1}$$

with N being the sample size of the (sub)-testing data set, λ_n the true value of observation n, $\hat{\lambda}_n$ the estimated value of observation n and $\overline{\lambda}$ being the sample mean of the (sub)-testing data set. Note that the best parameters obtained from the inner cross-validations may vary across different (outer) folds of training and test sets. Thus, after an algorithm has been chosen as per the outcome of the outer cross-validation, the best parameters are then chosen according to a normal one-stage cross-validation. We choose 5 folds for the outer cross-validation and 2 folds for the inner one.

For each of the three algorithms reported above, we are interested in a set of parameters that specify the algorithms. These parameters are given in Table E.6 with their names as per the documentation of the software package Scikit-learn.

Table E.6
Table of parameters for each algorithm used in the nested cross-validation procedure.

Algorithm	Parameters
NN	activation functions: {'relu', 'logistic'} solver: {'lbfgs', 'adam'} learning rate: {'constant', 'adaptive'}
KNN	n neighbors: {5,7,10} weights: {'uniform', 'distance'}
RF	n estimators: {3,5,7}

Appendix F. Simulation

Here we discuss some technical specifications characterizing our simulation model. The model was designed and implemented in accordance with the requirements of TBA Group. It is characterized by an eventlist functionality. The eventlist keeps track of all events that are scheduled in the future, ordered in time of occurrence with the earliest on top of the list. The first event from the list is read and a reaction is given to this event which might trigger another event, leading to it being scheduled and added to the eventlist. Every time an event is added, the eventlist is sorted again. An example of the eventlist mechanism is given in Fig. F.6.

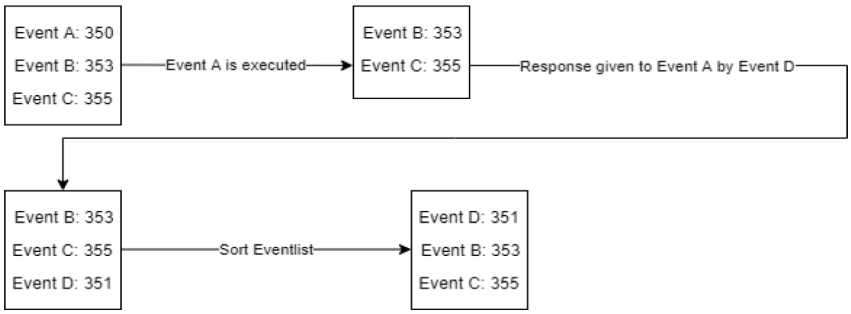


Fig. F.6. Illustration of eventlist mechanism.

When the vessel berth commences, and the quay cranes start discharging, we have that all trucks that serve the quay crane are lined up before it. Furthermore, we allow the quay crane to load and discharge containers before or even after the time epochs in which the containers were allocated according to the optimization problem (6)–(17) as long as there are trucks available for transporting the containers. This is because a quay crane is never forced to wait in practice. Similarly, a yard crane proceeds to processing a container order with the truck arrival, so that it does not deliberately delay container orders.

During discharge, a quay crane can begin discharging a container even when there is no truck in the queue. Let s_q be the time it takes a quay crane q to completely discharge a container and is shown by the bold line in Fig. F.7 shows withe bold line. Note that the horizontal distance is traversed twice; before and after fetching the container. We assume that the vertical distance to the truck

queue has deterministic travel time of 0.5 min. This is a reasonable assumption since there is little variation associated with these moves. Thus at time $s_q - 0.5$, shown by the dashed line in Fig. F.7, the quay crane checks whether there is an available truck. If there is, it proceeds with placing the container on the truck. Otherwise, it waits until a truck arrives. This waiting time is exclusive of the vertical travel time of 0.5 s.

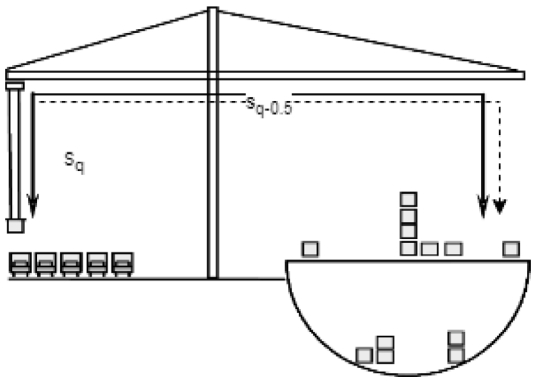


Fig. F.7. Quay crane service time illustration.

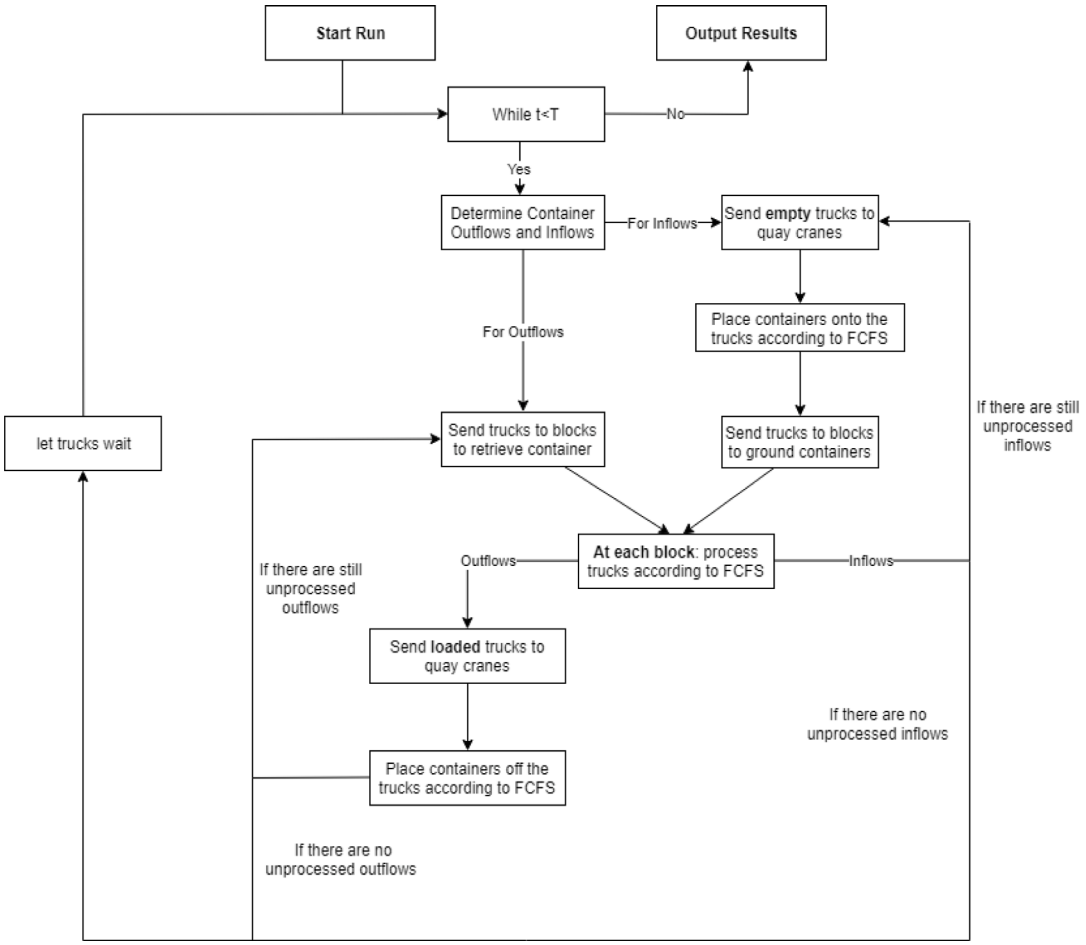


Fig. F.8. Flowchart depicting sequence of events that take place during simulation. Here T refers to the overall test period, which we take to be six weeks, and t is the current time.

We also allow trucks to depart from the block when delivering containers to a block – during discharge – before the containers are grounded which is closer to what happens in reality.[8] This means that a truck may leave the block once it is freed from the container. Just like with the quay cranes, we assume that the vertical travel distance is 0.5 min for the yard cranes so that trucks can leave after 0.5 min from entering service. A similar thing applies to trucks delivering containers to the quay cranes — during loading.

Before a quay crane starts loading, we require that every truck first returns to the quay crane without entering the queue before it proceeds to taking loading orders. This is because driving from some blocks to others is not allowed if that causes the trucks to drive in the opposite direction. Additionally, after a truck delivers a loading order at a quay crane and the quay crane's loading work queue is finished, the truck is parked next to the crane so as not to disrupt other trucks. The quay crane finishes discharging once a truck picks up the last container in its discharge work queue, and finishes loading once its loading work queue is finished **and** all its trucks return. After that, the quay crane is changed to idle state. When all the cranes of a given vessel become idle, the berth terminates and the ship proceeds to depart. A flowchart summarizing the simulation procedure is given in Fig. F.8.

Appendix G. Benchmark

For the first subproblem where we optimize the distribution of workloads among the blocks, we have the following decision variables. R_i^t representing the number of discharge containers assigned to block i in epoch t, A_t representing the maximum work queue per block among all blocks in epoch t and B_t representing the minimum one. The only parameters are L_i^t representing the loading orders for block i in period t and N^t representing the total number of containers discharged in epoch t. The subproblem is given by:

$$\min \sum_{t=1}^{T} [A_t - B_t] \tag{G.1}$$

$$s.t. \quad \sum_{i \in B} R_i^t = N^t \qquad 1 \le t \le T \tag{G.2}$$

$$A_t \ge R_i^t + L_i^t \qquad \forall i \in B, 1 \le t \le T \tag{G.3}$$

$$B_t \le R_i^t + L_i^t \qquad \forall i \in B, 1 \le t \le T \tag{G.4}$$

$$R_i^t \in \mathbb{Z}^+ \qquad \forall i \in B, 1 \le t \le T \tag{G.5}$$

$$0 \le A_t, B_t \le \Delta \times (\mu_y - \epsilon) \qquad 1 \le t \le T \tag{G.6}$$

Objective (G.1) minimizes the difference between the maximum and minimum work load among the blocks in each epoch t. Zhang et al. (2002) explain how this is an effective strategy to minimize waiting times at the yard. Constraint (G.2) ensure that every discharged container is allocated to a block. Constraints (G.3) and (G.4) define the maximum and minimum work load among all the blocks. Constraints (G.5) and (G.6) define the domains of the variables. We bounded the A_t and B_t variables from above by $\Delta \times (\mu_y - \epsilon)$ in analogy to Section 4.1, although the approach is generally oblivious of such working capacity constraints.

The next step is to decide on the block assignments. For this problem, the R_i^t values from problem (G.1)–(G.6) are fixed and taken as parameters. We refer to them as $\overline{R_i^t}$. The other parameters are the N_{qj}^t / N_q^t and t_{ji} values. Note that $N^t = \sum_{q \in Q} (N_q^t + \sum_{j \in L} N_{qj}^t)$ in the first subproblem. Our only decision variable is X_{qj}^{ti} which has a similar interpretation as R_{qj}^{ti} in Section 4.1. The second subproblem is given as follows:

$$\min \sum_{t=1}^{T} \sum_{i \in B} \sum_{q \in Q} \sum_{j \in L} t_{\hat{j}i} X_{qj}^{ti} \tag{G.7}$$

$$s.t \qquad \sum_{i \in B} X_{qj}^{ti} = N_{qj}^t \qquad 1 \le t \le T, \forall q \in Q, \forall j \in L \tag{G.8}$$

$$\sum_{i \in B} X_q^{ti} = N_q^t \qquad 1 \le t \le T, \forall q \in Q \tag{G.9}$$

$$\sum_{q \in Q} (X_q^{ti} + \sum_{j \in L} X_{qj}^{ti}) = \overline{R_i^t} \qquad 1 \le t \le T, \forall i \in B \tag{G.10}$$

$$X_{qj}^{ti}, X_q^{ti} \ge 0 \qquad 1 \le t \le T, \forall i \in B, \forall q \in Q, \forall j \in L \tag{G.11}$$

Objective (G.7) minimizes the driving time during discharge; recall that $\hat{j}$ is the location of q. Constraints (G.8) and (G.9) ensure that every discharged container per quay crane is allocated to a block. Constraint (G.10) ensures that the workloads of the blocks in each epoch are respected as per the optimization in (G.1)–(G.6). Since the integrality property is guaranteed for such an assignment problem, it suffices to impose a positive domain in constraint (G.11) on the decision variables.

[8] In Section 5.1, it was assumed that trucks have to stay until service completion.

Appendix H. Verifications

This section is dedicated to verifying the approximations introduced throughout this paper. We seek to give a clear idea about the estimation accuracy of the quay crane arrival rates in Appendix H.1 given dependent truck movements, the assumption of exponentially distributed truck inter-arrival times at the quay crane in Appendix H.2, estimated waiting times at the yard using the formula based on $M/M/1/K$ queuing systems in Appendix H.3. Lastly a brief account on the practicality of λ_q^t - based on $\hat{f}(.)$ - when some of the assumptions in Section 5 no longer hold is provided in Appendix H.4.

H.1. Quay crane arrival rate

This section deals with validating the estimated performance measures using Algorithm 4. The analysis is based on the assumption that trucks delays do not affect other trucks. We would like to test whether this assumption affects the results significantly. To do this, we run a simulation for a terminal with fixed truck allocations determined by some matrix A. The simulation is based on a setting with the strict assumptions of Section 3 except for the independence of truck movements; meaning that trucks cannot overtake each other as is the case in reality. We then estimate the simulated arrival rate for a set of cranes under the allocation A. The simulated estimates are then compared to the theoretical ones from Algorithm 4 for the same allocation. The process is repeated for several allocations.

Since, we are only interested in the quantity $\lambda_q(A)$, we check for this quantity only. Nonetheless, this could serve as an indication of the estimation accuracy of the other quantities; namely $E[w_i^q(A)]$, $E[w_q^i(A)]$ and $\lambda_q^i(A)$, since $\lambda_q(A)$ depends directly on these quantities. For a given A, the theoretical estimates can be obtained directly from Algorithm 4. The simulated estimates on the other hand are determined using:

$$\lambda_q = \frac{\overline{N_q(s)}}{s} \tag{H.1}$$

with $\overline{N_q(s)}$ being the number of counted truck arrivals at quay crane q during the course of s minutes of simulations. To model $s \to \infty$, we choose $s = 1,000,000$ min.

The result of the comparison between the simulated and theoretical estimates are given in the scatter plot in Fig. H.9. Each point corresponds to the theoretical estimate of the arrival rate on the x-axis, and the simulated one on the y-axis for a quay crane under some allocation A. The red line corresponds to the $y = x$ line, so that the closer the points are to this line, the closer the theoretical estimates are to the simulated ones. The figure confirms the accuracy of our estimates using Algorithm 4 as the majority of the point lie very closely if not on the line.

We can thus safely conclude that the assumption of independence of truck movements is not impractical in a setting where the truck delays actually affect each other while all other assumptions in Section 3 hold. Nonetheless, it does not answer the question of the practicality of using forecasts based on these assumptions in a setting where the many of them are no longer valid. Appendix H.4 answers this.

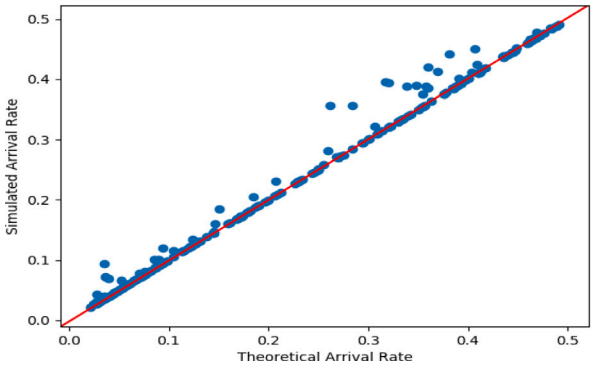


Fig. H.9. Scatter plot depicting simulated arrival rates in a setting as per the assumptions in Section 3 on the y-axis, and the theoretical arrival rates from Algorithm 4 on the x-axis for different quay cranes under different allocations. The red line is the $y = x$ line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

H.2. Poisson truck arrivals

In this section, we assess the assumption of Poisson truck arrivals proposed in Section 5.2. To do this, we compare the histogram representing inter-arrival times of the trucks at the quay cranes during the simulation run for one of the scenarios drawn up in Section 6.2 with a histogram representing a sample of exponentially distributed random variables with the same mean as that of the inter-arrival times observed during the simulation run. A very similar pattern of histograms was observed for other scenarios. The comparison is illustrated in Figs. H.10 and H.11.

The histograms more or less resemble each other. The histogram of simulated inter-arrival times captures the main features of the exponential distribution. Smaller values occur at a higher frequency than larger ones and the decline in frequency occurs at a

non-linear decreasing rate, although the values of the exponential distribution appear to be slightly more spread out. Nonetheless, we conclude that the assumption of Poisson truck arrivals is still reasonable, despite the normally distributed crane service times assumed in the simulation.

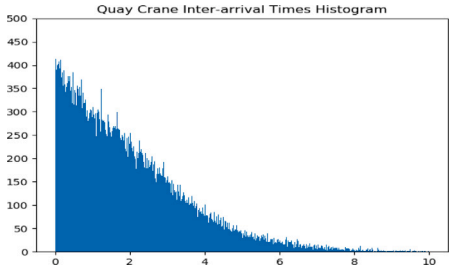


Fig. H.10. Histogram of truck inter-arrival times during the simulation run for Scenario 2 with $\alpha = 0.5$ and $n_q = 7$. The mean inter-arrival time was estimated to be 2.25.

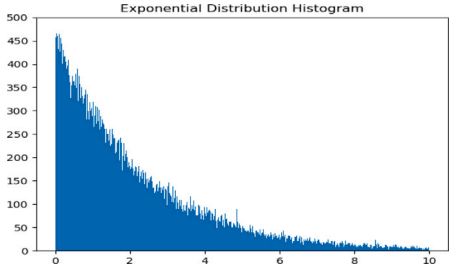


Fig. H.11. Histogram of a sample of random variable from an exponential distribution with mean 2.25.

H.3. Yard crane waiting times

Here we discuss the validity of estimating the expected waiting time $E[w_i(A)]$ at a block i using the formula of an $M/M/1/K$ queue. Recalling the notation in Section 5.1, we define $E[w_i(A)]$ as follows:

$$E[w_i(A)] = \frac{\sum_{q\in Q} V_i^q(A) E[w_i^q(A)]}{\sum_{q\in Q} V_i^q(A)} \tag{H.2}$$

with $E[w_i^q(A)]$ determined using Algorithm 4. For a given $\lambda_i(A) = \sum_{q\in Q} \lambda_q^i(A)$, we would like to see where the $M/M/1/K$ formula gives a reasonable estimate with $K = \sum_{q\in Q} V_i^q(A)$. Let this estimate be $W_{m/m/1/K}(\lambda_i(A))$. The comparison is carried out by generating a sample of allocations and determining the expected waiting time at a set of blocks under each allocation just like in Appendix H.1. Fig. H.12 depicts a scatter plot of $E[w_i(A)]$ on the x-axis against $W_{m/m/1/K}(\lambda_i(A))$ on y-axis, where each coordinate corresponds to a block i under some allocation A. The red line is the $y = x$ line. The estimates from the $M/M/1/K$ formula seem fairly reasonable for small values of $E[w_i(A)]$, although they are slightly larger. However, they are serious underestimates for larger values of $E[w_i(A)]$.

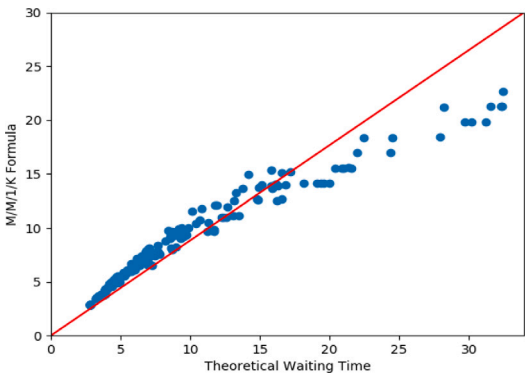


Fig. H.12. A scatter of $E[w_i(A)]$ on the x-axis against $W_{m/m/1/K}(\lambda_i(A))$ on the y-axis for different blocks under different allocations A. The red line is the $y = x$ line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Indeed, it appears that the forecast error expressed as a percentage of $E[w_i(A)]$ which is calculated by:

$$\frac{(E[w_i(A)] - W_{m/m/1/K}(\lambda_i(A))) \times 100}{E[w_i(A)]} \tag{H.3}$$

decreases to values around −10% as the arrival rate $\lambda_i(A)$ at the block increases from 0 onward before rising up to over 30% as $\lambda_i(A)$ approaches μ_y.[9] This pattern can be observed in Fig. H.13 which plots the percentage forecast error in (H.3) on the y-axis against $\lambda_i(A)$ on the x-axis for the corresponding block under a given allocation A. Nonetheless, one need not worry about accepting solutions with very large underestimated $E[w_i(A)]$ values as the $M/M/1/K$ formula – despite its underestimation – still results in large estimates of the expected waiting times at the blocks for such high arrival rates $\lambda_i(A)$. Consequently, our meta-heuristic is likely to reject these solutions which encompass rather low arrival rates at some of the quay cranes, due to the long waiting times at the blocks, possibly violating Constraint (19).

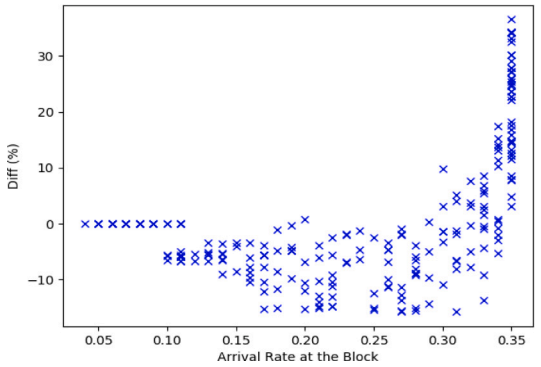


Fig. H.13. A plot of the percentage forecast error in (H.3) on the y-axis against the arrival rate $\lambda_i(A)$ at a block i under allocation some allocation A on the x-axis.

H.4. Practicality of theoretical arrival rates

This section deals with the most imperative questions regarding "How accurate are our λ_q^t values given that many our initial assumptions in Section 3 do not hold anymore?". To answer this question, we generate a list of discharge work queues for an arbitrary number of cranes over a period of 3000 min and compare the forecasted arrival rate λ_q^t with the actual arrival rate in the simulation during the course of epoch t for an arbitrary quay crane. We considered two settings where one has $M = 3$, $n_q = 7$ and the other $M = 4$, $n_q = 9$.

The deductions from our comparison show that our λ_q^t values do **not** in general reflect the actual arrival rate in epoch t which was calculated in a similar fashion as in (H.1) except that now s lasts for Δ minutes only. The results from the first setting are shown in Fig. H.14 where the blue and red lines represent λ_q^t and the simulated arrival rates and Fig. H.15 which depicts a scatter plot of both arrival rates. Fig. H.14 shows that our estimates consistently undermine the regular arrival rates,[10] which we expected in Section 4.2 due to the over estimation of the service times at either cranes by assuming that the trucks have to wait until the container is grounded by the yard crane and that the quay crane does not initiate a discharge operation before a truck arrival. Furthermore, the estimates do not exactly correlate with the actual arrival rates as can be seen from the cloud of scatters in Fig. H.15 which give a small correlation of 0.07.

A similar thing could be said about the second setting with results illustrated in Figs. H.16 and H.17. Here, although the problem of under-estimating arrival rates appears to be less relevant with the inclusion of more trucks — that partially compensate the effect of over-estimated waiting times, the correlation with the actual arrival rates declines to −0.05. Since the correlations assume such small values in absolute terms, we are inclined to conclude that there is no correlation with our forecasted arrival rates and the actual ones.

There are several explanations that could justify the absence of a correlation. Firstly, the simulated arrival rates appear to be much more variable as can be seen in Figs. H.14 and H.16, attaining both much higher and lower values. The latter could be due to under-estimating the waiting times at the blocks when truck arrivals are very frequent – which even counters the over-estimation of the service times – as explained in Appendix H.3. A second factor could be the presence of a time-lag, meaning that the arrival rate of the quay crane takes some time to converge to the forecasted value, although the length of this lag is not clear to us. Thirdly, and with regards to time-convergence, the concept of an arrival rate is based on a long-run average. Estimating this quantity based on the number of occurrences that occurred in a short time interval of $\Delta = 15$ min as in our case is likely to result in deviations from the expected theoretical value. Despite these shortcomings, the most pressing factor is the ability to generate reductions in the berth times of the vessels which our model does anyway.

[9] In our case, we had that $\mu_y = \frac{6}{17} \approx 0.353$.

[10] This explains the decrement d in (19).

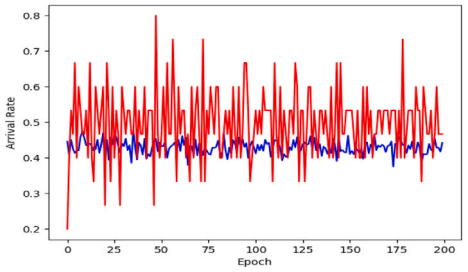


Fig. H.14. Graph comparing the forecasted arrival rates λ_q^t in blue with the ones retained from the simulation in red in each epoch for $M = 3$, $n_q = 7$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

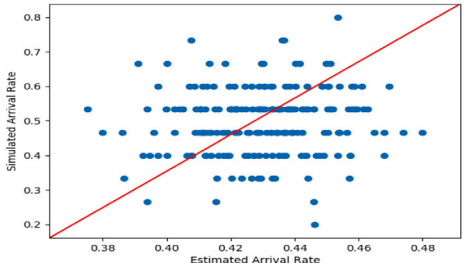


Fig. H.15. Scatter plot of the simulated arrival rates against the forecasted ones for $M = 3$, $n_q = 7$.

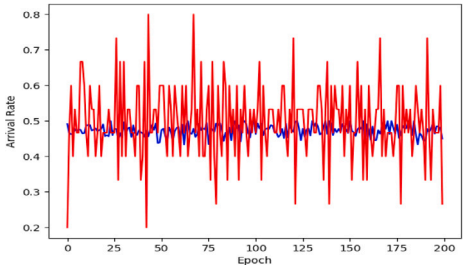


Fig. H.16. Graph comparing the forecasted arrival rates λ_q^t in blue with the ones retained from the simulation in red in each epoch for $M = 4$, $n_q = 9$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

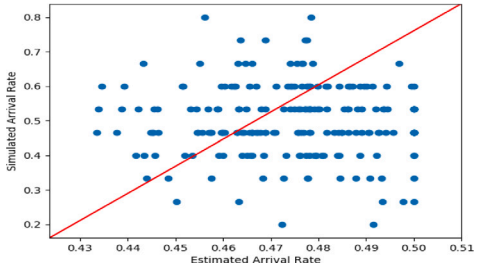


Fig. H.17. Scatter plot of the simulated arrival rates against the forecasted ones for $M = 4$, $n_q = 9$.

Appendix I. Value of the probabilistic model

This section analyzes the validity of the probabilistic model. In particular, we investigate whether it is possible to achieve further reductions in the mean berth time without performance measures constraints on the discharge process. This simply boils down to minimizing driving distances during loading by placing the containers as close as possible to their future pick up locations. To do that, we make use of the disjoint optimization of Zhang et al. (2003), except that we replace term $t_{\hat{j}i}$ with t_{ij} in the objective function (G.7). We compare the resulting mean berth times with those from Zhang et al. (2003) and our heuristic. The results are reported in Tables I.7–I.9 for the three scenarios with different truck numbers. Note that α is omitted since the disjoint optimization

is independent of the parameter. We refer to this model as the No Probabilistic Component (NPC) model. Here 'Mean' refers to the mean berth time, 'Decrease NPC' refers to the percentage decrease in the mean berth time under the NPC policy with respect to the (Zhang et al., 2003) and 'Decrease PC' indicates the highest percentage decrease in the mean berth time – compared to Zhang et al. (2003) – using our heuristic with some value of α for the same scenario in Section 6.2.

Table I.7
Results of Scenario 1, characterized by $\Gamma = \frac{1}{600}$, $C = 25000$, $\xi = 3$ and $M = 3$ with 105 vessel arrivals over a period of six weeks.

n_q	Mean	Decrease NPC (%)	Decrease PC (%)
7	380.31	1.02%	2.23%
8	376.72	0.42%	1.3%
9	375.47	0.21%	0.76%

Table I.8
Results of Scenario 2, characterized by $\Gamma = \frac{1}{480}$, $C = 30000$, $\xi = 5$, $M = 3$ with 128 vessel arrivals over a period of six weeks.

n_q	Mean	Decrease NPC (%)	Decrease PC (%)
7	375.73	1.18%	2.44%
8	374.10	−0.09%	1.25%
9	371.73	0.01%	0.68%

Table I.9
Results of Scenario 3, characterized by $\Gamma = \frac{1}{360}$, $C = 40000$, $\xi = 7$, $M = 4$ with 179 vessel arrivals over a period of six weeks.

n_q	Mean	Decrease NPC (%)	Decrease PC (%)
7	331.65	−0.28%	1.86%
8	328.67	−1.19%	0.95%
9	326.66	−1.27%	0.52%

The results clearly illustrate the value of the probabilistic model due to the higher percentage decrease in mean berth time in column 'Decrease PC' across all scenarios and cases. It appears that grounding the containers as close as possible to their loading locations is not a lucrative strategy when the OYD is very high as in Scenario 3, leading to not only a decline in the percentage decrease compared to our heuristic, but also compared to Zhang et al. (2003) as demonstrated by the negative values in column 'Decrease NPC'. Consequently, we can confidently nominate the probabilistic model as a useful tool for optimization.

Appendix J. Supplemental figures

In this section, we provide some of the figures used to adhere the findings in Section 6.2.

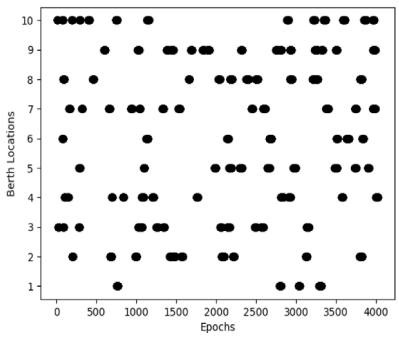


Fig. J.18. Scenario 1 vessel schedule: low traffic density.

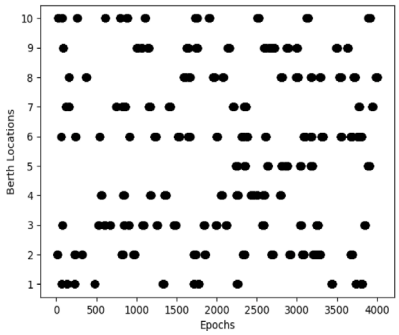


Fig. J.19. Scenario 2 vessel schedule: medium traffic density.

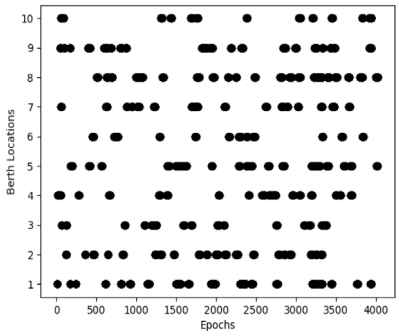


Fig. J.20. Scenario 3 vessel schedule: high traffic density.

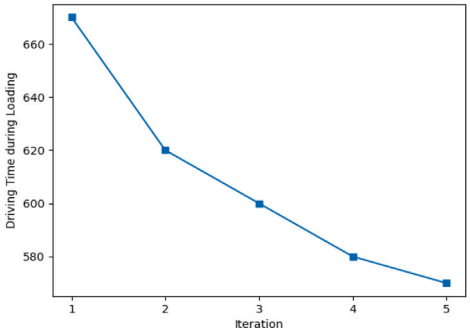


Fig. J.21. Lowest attained objective value of (6) over the course of iterations for our meta-heuristic.

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